The center and the ends of the elephant random walk with rare reversals

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Abstract

In the elephant random walk each step copies a uniformly chosen earlier step and reverses it with probability ε. We let ε tend to zero as the number of steps N grows, with $N\varepsilon$ of order $\log N$. The number of steps that disagree with the first step is approximated by a truncated Luria–Delbrück law, with an explicit total variation bound. Its Landau local limit has an explicit rate and locates every mode near one of the two ends. The same Landau law arises from the fixed-memory superdiffusive limit as the memory tends to one. For even N, the origin and each end cross exactly once, at an expected reversal count of $2\log N - \log\log N - \log 16 + o(1)$. We bound the relative error of the origin probability $4\varepsilon/(N+2)$. An exact sign argument proves that, at the crossing, the origin is the unique interior minimum for every even $N \geq 8$. The law has exactly two symmetric intervals of maximizing bins, with strict increase from the ends and strict decrease toward the center. This uses certified endpoint comparisons at finitely many lengths. For a reversible fluctuation assay with one sensitive founder and symmetric switching at birth, the total variation bound controls the error of the compound Poisson experiment.

Keywords: elephant random walk; random recursive tree; broadcasting on trees; compound Poisson approximation; Landau distribution; computer-assisted proof.
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1 Introduction

In the elephant random walk of Schütz and Trimper [1], each step copies a uniformly chosen earlier step and reverses its sign with probability ε. The first step is fair. Put $x = N\varepsilon$. We study $x \asymp \log N$, so reversals become rarer even though their expected number $x - \varepsilon$ grows. Most limit theorems instead fix ε [4, 5, 6, 7, 8].

The number of steps disagreeing with the first is the resistant-cell count in Bokes, Hlubinová and Singh's reversible fluctuation assay [19], with one sensitive founder, uniform parent choice and equal switching probabilities at birth. Their first-switch and Pólya-urn calculation already gives the first-order center coefficient $4/(N+2)$ [19, Eqs. (6)–(10)], and their random-start approximation uses two Landau densities [19, Eqs. (58)–(60)]. Our contribution is quantitative, namely an explicit total variation bound for the truncated Luria–Delbrück approximation, a Landau local limit theorem with a rate, and exact crossing and shape results. The approximation also gives error bounds for a constrained likelihood estimator in that assay (Theorem S4.1 of Online Resource 1).

A weak limit connects our Landau law to the classical superdiffusive law as the memory tends to one (Corollary 4.1).

For even N, when is the origin as likely as each endpoint $\pm N$? Near $\varepsilon = 0$ the endpoints win, while at $\varepsilon = \frac{1}{2}$ the steps are independent and the origin wins. We prove that they cross exactly once and locate the crossing at $x = 2\log N - \log\log N - \log 16 + o(1)$. The persistent walk, which copies the last step rather than a uniform earlier step, crosses at about $\log N$ switches [10]. That comparison originates in Kagey's Galton-board problem [9], so we call the possible counts of right steps bins.

For comparison of conventions, the original memory parameter is $1-\varepsilon$, the parameter a of [7] is $1-2\varepsilon$, and the broadcasting flip probability is ε. The broadcasting tree representation is recalled in Lemma 3.1 [2, 3].

1.1 Setting and notation

The walk takes N steps $X_1, \dots, X_N \in \{+1, -1\}$, where $+1$ means right. Let $A_N = \#\{i \le N : X_i = +1\}$ be the number of right steps, and write $P_N(k) = \mathbb{P}(A_N = k)$ for the probability of bin k. Unless we say otherwise the first step is fair, $\mathbb{P}(X_1 = +1) = \frac{1}{2}$, and the rules treat the two directions alike, so $P_N(k) = P_N(N-k)$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for probability and expectation given $X_1 = +1$. For even N we put $h = N/2$ and call $C_N = P_N(h)$ the center and $E_N = P_N(N) = P_N(0)$ the end. Whenever C_N appears, N is even. The crossing is the value of the parameter at which $C_N = E_N$. All logarithms are natural, $L = \log N$, $H_k = \sum_{j \le k} 1/j$ is the k-th harmonic number, and γ is Euler's constant. Limits, $O(\cdot)$ and $o(\cdot)$ refer to $N \to \infty$, over even N when C_N appears. The constants implied by $O(\cdot)$ are absolute unless we name what they depend on.

The elephant walk (Definition 2.1) has the reversal probability $\varepsilon \in (0,1)$ and $x = N\varepsilon$. Its crossing is ε_N, and $x_N = N\varepsilon_N$. The symbols that only one section uses are defined in that section, where they first appear.

1.2 Main results

Theorem 1.2 is the main approximation tool. We first state the center and crossing bounds, then the bulk and shape results.

Theorem 1.1 (the center and the crossing)**.** *Let $N \ge 2$ be even.*

(a) The ratio C_N/E_N is strictly increasing in $\varepsilon \in (0,1)$, and the crossing ε_N exists, is unique and lies in $(0, \frac{1}{2})$ (Proposition 2.3).

(b) For $0 < \varepsilon < \frac{1}{2}$ we have $C_N \ge \frac{4\varepsilon}{N+2}(1-2\varepsilon)$. If $\varepsilon(1+\log N) \le \frac{1}{100}$, then $C_N \le \frac{4\varepsilon}{N+2}\, N^{2\varepsilon} e^{5200\varepsilon}$ (Theorem 2.6, whose constant 5200 uses an enclosure in ball arithmetic).

(c) As $N \to \infty$ over even N, the crossing satisfies $x_N + \log x_N = 2\log N - \log 8 + O(\log^2 N/N)$, and so

$$x_N = 2\log N - \log\log N - \log 16 + O\Big(\frac{\log\log N}{\log N}\Big)$$

(Theorem 2.7).

So $C_N = \frac{4\varepsilon}{N+2}(1 + O(\varepsilon \log N))$ when $\varepsilon \log N$ is small, and the error factor is $1 + O(\log^2 N/N)$ at the crossing. For the persistent walk with switch probability q, the crossing has $T = Nq$ close to $\log N$, and Theorem S1.7 of Online Resource 1 compares the two crossings to the next order.

For the rest of the law let M_N be the number of steps that disagree with the first step. Its law $f_N(m) = \mathbb{P}(M_N = m)$ does not depend on X_1, and $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$ (Lemma 2.2). Let CP_N be the law of $\sum_{d=1}^{N-1} d\,\mathcal{N}_d$, where the $\mathcal{N}_d$ are independent Poisson variables with means $x/(d(d+1))$. Put $\lambda_m = (m - x\log x)/x$, and let

$$f_L(\lambda) = \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\lambda t} e^{B(t)}\,dt, \qquad B(t) = -it\log|t| - \frac{\pi}{2}|t|,$$

be the Landau density.

Theorem 1.2 (the bulk)**.** *Suppose $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Uniformly over integers $m \geq 0$,*

$$\bigl|x f_N(m) - f_L(\lambda_m)\bigr| = O\Bigl(\frac{1+\log x}{x} + \frac{x^3\log^2 N}{N}\Bigr),$$

and $(M_N - x\log x)/x$ converges in law to the density f_L. Theorems 3.8 and 3.9 give explicit finite-N bounds. The density has a unique mode $\lambda_{\mathrm{L}} \approx -0.222783$ (Lemma 3.10, computer-assisted). For $\log N \leq x \leq 2\log N$ and all sufficiently large N, every mode $a^* \geq N/2$ of P_N satisfies

$$\bigl|N - a^* - x\log x - \lambda_{\mathrm{L}} x\bigr| \leq K\sqrt{x\log x},$$

and the same bound holds for every mode of f_N in place of $N - a^*$ (Corollary 3.11). We do not compute K.

At the crossing each hump is about $x\log x - 0.2228\,x$ bins from its end. For the shape between them put $D_N(a) = P_N(a+1) - 2P_N(a) + P_N(a-1)$ and $m^* = N - a^*$ for a mode $a^* \geq N/2$.

Theorem 1.3 (the shape)**.** *(a) For $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$, delete zeros from the signs of D_N and merge equal neighbors. The resulting pattern is a subsequence of $(+,-,+,-,+)$ (Theorem 3.16).*

(b) For every even $N \geq 8$, the crossing law has exactly two symmetric intervals of maximizing bins. It strictly rises to each interval and strictly falls from it toward the center, its unique interior minimum (Corollary 3.19). This uses the finite endpoint-dip certificates of Proposition 3.7. The whole statement, certificates included, is kernel-checked in Lean (Section 5).

(c) For all sufficiently large N, even or odd, uniformly for $\log N \leq x \leq 2\log N$, the law is strictly convex between the humps (Theorem 3.23). The threshold is not computed. Certified computations give that prescribed convexity window at the crossing for every $5 \leq N \leq 2000$ (Proposition 3.24).

Only the prescribed convexity window of part (c) remains open between the finite range and the eventual threshold (Conjecture 3.26). The minimum and two-hump conclusion of part (b) holds throughout.

The three computational inputs are described in Section 5, with the result each one supports. The tables and figures illustrate the theorems and are not used in their proofs.

Why twice as many reversals? An endpoint has mass about $\frac{1}{2}e^{-x}$. In the persistent walk, let q be the switch probability and $T = Nq$. Its runs have typical length N/T, and the right-step count spreads over about $N/\sqrt{T}$ bins. Its central mass is of order $\sqrt{T}/N$, which balances e^{-T} at $T \approx \log N$. An elephant reversal instead starts a family of descendants, governed by Pólya's urn. Reaching the middle requires an early reversal whose family occupies about half the walk. This costs order ε, and the exact middle costs another $1/N$. Thus the center has mass about $4\varepsilon/N = 4x/N^2$, balancing the endpoint at $x \approx 2\log N$. Most of the roughly $2\log N$ reversals there occur too late to produce a large family.

1.3 Prior work

Fixed-parameter limits for the elephant walk are well developed [4, 5, 6]. Guérin, Laulin, Raschel and Simon prove conditional unimodality [7, Section 2, proof of Theorem 1.1]. Trinas and Valle's superdiffusive local limit and study of log-concavity [8, Theorem 3.1 and Eq. (4.10)] also fix the parameter. Engländer and Volkov's coin-turning walk instead switches at time n with probability p_n. For $p_n = a/n$ eventually, its occupation fraction tends to Beta(a, a) [13, Section 4, Theorem 1]. We vary the reversal probability with the final length, not with time within one walk.

Kürsten's percolation coupling [11] copies a parent with probability $1 - 2\varepsilon$ and otherwise gives the new vertex a fair sign. Baur and Bertoin [14] give the urn connection. Our regime is strongly supercritical in Baur's classification [12], following Bertoin [15]. These works describe cluster sizes; Baur's Corollary 1 gives a Poisson limit for the largest nonroot clusters. Miyazaki and Takei obtain related root-cluster limits via the minimal elephant walk [16, Theorems 2.3, 2.4 and Section 2.3]. They do not estimate the exact central mass in our regime.

Without truncation, our compound Poisson law is the Lea–Coulson Luria–Delbrück law. Kessler and Levine describe its small-parameter profile and Landau limit [17, Eqs. (14), (18), (19)], as well as the dip threshold and mode near $\mu N(\log \mu N - 0.22)$ [17, Section 2]. Möhle proves the stable limit and density formulas [20, Theorem 4.1 and Corollary 4.1]; his Corollary 4.2 gives a uniform lattice local limit for integer parameters, without a rate. Schweinsberg [21] and Bertoin [22] obtain the same stable law for root-cluster fluctuations at retention $1 - c/\log n$. We add truncation at $N - 1$, an explicit local rate and transfer to the elephant law, from which its dip and mode estimates follow.

Broadcasting papers study reconstruction of the root color from the terminal tree [2, 3]; color-count limits and martingale error bounds are also available [23, 24]. Here the center is the probability of a color tie, relevant to majority-vote error, with the flip probability tending to zero. Gut and Stadtmüller's count [25, (1.5)] has next-step probability pk/n, whereas ours has $\varepsilon + (1 - 2\varepsilon)k/n$, so their chain cannot leave zero. The irreversible mutation expansion of Kessler and Levine [18, Eqs. (10), (13), (14)] has the same first derivative at $\varepsilon = 0$ as our law, but not our uniform relative bound at $N\varepsilon \asymp \log N$. We found no such central estimate in the literature reviewed.

1.4 Organization

Section 2 proves the crossing theorem, using the center bounds of Appendix A. Section 3 treats the bulk, modes and convexity. The compound Poisson and Landau proofs are in Appendix C. Appendix B states the auxiliary family estimates, Appendix D proves them, and Appendix E proves the second-difference lemmas. Section 4 gives the fixed-memory limit. Section 5 describes the numerical and formal checks. Online Resource 1 contains the second-order crossing, tail and odd-N results and the likelihood bound for the fluctuation assay. Every proof used by the main theorems is in the article.

2 The elephant walk and its crossing

We reduce the elephant walk to a counting chain and prove that the crossing is unique. We then compute the center to first order in ε, bound it from both sides, and locate the crossing.

Definition 2.1 (elephant walk)**.** Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with

probability ε. All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As in Section 1.1, $A_N = \#\{i \le N : X_i = +1\}$ and $P_N(k) = \mathbb{P}(A_N = k)$. The crossing ε_N is the value of ε with $C_N = E_N$ (it exists and is unique by Proposition 2.3), and $x_N = N\varepsilon_N$.

A flip is what the introduction called a reversal. Given the first step, all that matters is how many steps disagree with it. So let M_t be the number of $i \le t$ with $X_i \ne X_1$, and put

$$r_t(b) = \varepsilon + (1-2\varepsilon)\frac{b}{t}, \qquad f_N(m) = \mathbb{P}_+(M_N = m).$$

Lemma 2.2. *The process $(M_t)_{t\ge1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b+1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1. For $0 \le k \le N$, $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)]$. In particular $C_N = f_N(h)$, $E_N = \frac{1}{2}(1-\varepsilon)^{N-1}$, and $P_N(k) = P_N(N-k)$.*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t. So $X_{t+1} \ne X_1$ with conditional probability $(1-\varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1, which proves the first claim. On $\{X_1 = +1\}$ we have $A_N = N - M_N$, and on $\{X_1 = -1\}$ we have $A_N = M_N$. Averaging over X_1 gives the formula for $P_N(k)$, which does not change under $k \mapsto N-k$. Finally $M_N \le N-1$, so $f_N(N) = 0$, and $f_N(0) = \prod_{t=1}^{N-1}(1 - r_t(0)) = (1-\varepsilon)^{N-1}$. □

So the end is that of the persistent walk with switch probability ε. More flips always raise the center relative to the end.

Proposition 2.3. *Let $N \ge 2$ and $1 \le k \le N-1$. Then $f_N(k)/f_N(0)$ and $P_N(k)/E_N$ are strictly increasing in $\varepsilon \in (0,1)$. In particular C_N/E_N is strictly increasing, the crossing ε_N exists and is unique, and $\varepsilon_N < \frac{1}{2}$. Also $P_N(k) \ge 2E_N$ for $\varepsilon \ge \frac{1}{2}$, and in particular $C_N \ge 2E_N$ there.*

Proof. We compare each path of M with the path that never flips, one step at a time. The probability of a path of M is a product of $N-1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s. The path that never flips has probability $f_N(0) = (1-\varepsilon)^{N-1}$, which is one factor $1-\varepsilon$ for each step. So we divide each factor by $1-\varepsilon$. With $y = b_s/s \in [0,1)$ the two kinds of factors become $\frac{\varepsilon+(1-2\varepsilon)y}{1-\varepsilon}$ and $1 - y\frac{1-2\varepsilon}{1-\varepsilon}$, whose derivatives in ε are $\frac{1-y}{(1-\varepsilon)^2} > 0$ and $\frac{y}{(1-\varepsilon)^2} \ge 0$. Both factors are positive, since $r_s(b_s) = \varepsilon(1-y) + (1-\varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε, and it is strictly increasing if the path has an up step. Summing over the paths with $k \ge 1$ up steps proves the claim for $f_N(k)/f_N(0)$. By Lemma 2.2, $P_N(k)/E_N = [f_N(k) + f_N(N-k)]/f_N(0)$, and both k and $N-k$ are at least 1. So the claim for $P_N(k)/E_N$ follows.

It remains to locate the crossing. The ratio C_N/E_N is continuous and strictly increasing in ε. As $\varepsilon \to 0$ it tends to 0, since $C_N \to 0$ (at $\varepsilon = 0$ the chain never leaves 0) and $E_N \to \frac{1}{2}$. For $\varepsilon \ge \frac{1}{2}$ every $r_s(b)$ and every $1 - r_s(b)$ lies in $[1-\varepsilon, \varepsilon]$, so each path of M has probability at least $(1-\varepsilon)^{N-1} = 2E_N$. Every m with $0 \le m \le N-1$ is the end of some path, so $f_N(m) \ge 2E_N$. Both k and $N-k$ lie in $\{1, \dots, N-1\}$, so $P_N(k) = \frac{1}{2}[f_N(N-k) + f_N(k)] \ge 2E_N$, and in particular $C_N \ge 2E_N$ there. So the ratio equals 1 at exactly one ε, and this ε lies in $(0, \frac{1}{2})$. □

2.1 First order, one early flip

When $\varepsilon = 0$ the chain M never leaves 0. After a first flip, and to first order in ε, it is Pólya's urn.

Lemma 2.4 (Pólya's urn)**.** *Let $2 \le v \le N$. At time v an urn holds one ball of one color and $v-1$ of the other. At each time $k = v, \dots, N-1$ we draw a ball uniformly and return it with*

one more ball of its color. Let $\beta_v(d)$ be the probability that at time N there are d balls of the first color. Then $\beta_v(d) = \binom{N-d-1}{v-2}/\binom{N-1}{v-1}$ for $1 \le d \le N-v+1$, and $\beta_v(d) = 0$ for other d. Also β_v is nonincreasing on $\{1, \dots, N-v+1\}$ and $\sum_d d\,\beta_v(d) = N/v$. For $1 \le d \le N-1$,

$$\sum_{v=2}^{N} \beta_v(d) = \frac{N}{d(d+1)}, \qquad \sum_{v=2}^{N} (v-2)\,\beta_v(d) = \frac{2N(N-d-1)}{d(d+1)(d+2)}.$$

Proof. We compute the probability of one order of the draws and count the orders. At time k the urn holds k balls. So if s of them are of the first color, the next draw is of that color with probability s/k. Take one order of the $N-v$ draws in which i draws are of the first color. The draws of the first color contribute the numerators $1, 2, \dots, i$, the other draws contribute $v-1, v, \dots, N-i-2$, and the denominators are $v, v+1, \dots, N-1$. So this order has probability $\frac{i!\,(v-1)v\cdots(N-i-2)}{v(v+1)\cdots(N-1)} = \frac{(v-1)\,i!\,(N-i-2)!}{(N-1)!}$, whatever the order is. There are $\binom{N-v}{i}$ orders. With $d = i+1$ this gives $\beta_v(d) = a_d(v-1)\binom{N-v}{d-1}$ with $a_d = \frac{(d-1)!(N-d-1)!}{(N-1)!}$, and writing out the factorials shows that this is $\binom{N-d-1}{v-2}/\binom{N-1}{v-1}$. From the last form, $\beta_v(d)/\beta_v(d-1) = 1 - \frac{v-2}{N-d} \le 1$ for $2 \le d \le N-v+1$, so β_v is nonincreasing on its support. The fraction of balls of the first color is a martingale that starts at $1/v$. So the mean number of such balls at time N is N/v.

For the two sums we use the first form of $\beta_v(d)$ and put $i = N-v$, so that $v-1 = \binom{N-1-i}{1}$ and $(v-1)(v-2) = 2\binom{N-1-i}{2}$. The Chu–Vandermonde identity says that $\sum_i \binom{i}{a}\binom{N-1-i}{b} = \binom{N}{a+b+1}$ (sort the subsets of $\{1, \dots, N\}$ of size $a+b+1$ by their element number $a+1$, which we call $i+1$). With $a = d-1$ and $b = 1, 2$ it shows that the two sums are $a_d\binom{N}{d+1}$ and $2a_d\binom{N}{d+2}$, and these are the values in the statement. □

We need some more notation. Let $V_s(b)$ be the chance of ending at the center from state b at time s. We define it by the backward recursion $V_N(b) = \mathbf{1}\{b = h\}$ and $V_s(b) = r_s(b)V_{s+1}(b+1) + (1 - r_s(b))V_{s+1}(b)$ for $s < N$. So $V_s(b) = \mathbb{P}_+(M_N = h \mid M_s = b)$ whenever $\mathbb{P}_+(M_s = b) > 0$, and $V_s(b)$ is a polynomial in ε. In particular it is defined at $\varepsilon = 0$, without conditioning on an event of probability zero. For $1 \le t \le h$ let

$$\pi_t = \beta_{t+1}(h) = \frac{t}{N-t}\prod_{i=1}^{t-1}\frac{h-i}{N-i}, \qquad c_1 = \sum_{t=1}^{h}\pi_t, \qquad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \le t \le h}$ is a probability vector.

The first-order coefficient in part (b) below is known. With both switching probabilities equal to ε and one sensitive cell at the start, the number of resistant cells in the reversible fluctuation assay of Bokes, Hlubinová and Singh [19, Section 2 and Eq. (1)] is our chain M. They condition on the first switch and use Pólya's urn, as we do, and their first-order formula [19, Eqs. (6)–(10)] gives $\varepsilon N/(m(m+1))$ as the part of $f_N(m)$ of first order in ε, which is $\frac{4\varepsilon}{N+2}$ at $m = h$. We include the short proof because the decomposition (a) and the weights (c) are what the uniform bounds of Theorem 2.6 and Appendix A start from.

Proposition 2.5 (first order, as in [19, Eqs. (6)–(10)])**.** *Let $N \ge 2$ be even.*

(a) The center is $C_N = \sum_{t=1}^{h} \varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$.

(b) The center C_N is a polynomial in ε with $C_N(0) = 0$ (extended as a polynomial to $\varepsilon = 0$). At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C_N'(0) = c_1 = \frac{4}{N+2}$ exactly.

(c) The weights satisfy $\sum_t w_t t = 3 - \frac{6}{h+2}$, and $w_t \le 2t2^{-t}$ for $1 \le t \le h$.

Proof. (a) We condition on the time of the first flip. Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \le t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. A path with no flip ends at $0 \neq h$. So C_N is the sum of $\varepsilon(1-\varepsilon)^{t-1}V_{t+1}(1)$ over $1 \le t \le N-1$. Only $t \le h$ contribute. Indeed at time $t+1$ there are t steps equal to X_1, this number never decreases, and at the center it is h. So $V_{t+1}(1) = 0$ for $t > h$.

(b) Every transition probability is affine in ε, so C_N and $V_{t+1}(1)$ are polynomials in ε. By (a), $C_N(0) = 0$ and $C_N'(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$. So from time $t+1$ on, M counts the balls of the first color in the urn of Lemma 2.4 with $v = t+1$, and $V_{t+1}(1) = \beta_{t+1}(h) = \pi_t$. Since $\beta_v(h) = 0$ for $v > h+1$, Lemma 2.4 with $d = h$ gives $c_1 = \sum_{v=2}^{N} \beta_v(h) = \frac{N}{h(h+1)} = \frac{4}{N+2}$.

(c) In the same way the second sum of Lemma 2.4, with $d = h$ and $v = t+1$, gives $\sum_t (t-1)\pi_t = \frac{2N(h-1)}{h(h+1)(h+2)}$. Dividing by $c_1 = \frac{N}{h(h+1)}$ gives $\sum_t w_t(t-1) = \frac{2(h-1)}{h+2} = 2 - \frac{6}{h+2}$, and adding $\sum_t w_t = 1$ gives the first claim. For the second claim, each factor $\frac{h-i}{N-i}$ in π_t is at most $\frac{1}{2}$, so $\pi_t \le \frac{t}{N-t}2^{1-t}$. Hence $w_t = \frac{N+2}{4}\pi_t \le \frac{N+2}{2(N-t)}\, t2^{-t}$. For $t \le h$ we have $N - t \ge h$ and $\frac{N+2}{2h} = 1 + \frac{2}{N} \le 2$, so $w_t \le 2t2^{-t}$. □

So to first order the walk flips once, at some step $t+1$, and then runs as Pólya's urn from t balls of one color and one of the other. As $N \to \infty$ the final fraction of the color that started with one ball tends in law to a Beta$(1,t)$ variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$, and $\sum_t t2^{1-t} = 4$ explains the constant $4/(N+2)$. Since $w_t \le 2t2^{-t}$, only an early flip counts in this first-order term.

2.2 Two-sided bounds on the center

This subsection bounds the center from both sides. The bounds give the crossing in the next subsection.

Theorem 2.6. *Let $N \ge 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{2}$, $C_N \ge \frac{4\varepsilon}{N+2}(1-2\varepsilon)$.

(b) The bound $C_N \le \frac{4\varepsilon}{N+2}\, N^{2\varepsilon}e^{5200\varepsilon}$ holds whenever $0 < \varepsilon(1+\log N) \le \frac{1}{100}$.

In particular $\log(1-2\varepsilon) \le \log C_N - \log\frac{4\varepsilon}{N+2} \le \varepsilon(2\log N + 5200)$ in the range of (b).

Part (a) also holds for $\frac{1}{2} \le \varepsilon < 1$, trivially, because its right side is not positive there.

Idea of the proof. The proof is in Appendix A. After the first flip, M at time s is Pólya's urn with αs extra balls of each color, where $\alpha = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). They pull the minority fraction toward $\frac{1}{2}$. For (a) we drop them. For (b) we compare M with an urn with $\kappa(s+10)$ extra balls, where $\kappa = 2\varepsilon/(1-4\varepsilon)$. One step of M at time s multiplies the hitting probability of this urn by a factor of at most about $e^{\kappa/s}$, and the product of these factors is about $e^{\kappa H_N} \approx N^{2\varepsilon}$.

2.3 The crossing

By Theorem 2.6, $C_N \approx 4x/N^2$ when $\varepsilon = x/N$ and x is of order $\log N$. Also $E_N = \frac{1}{2}(1-x/N)^{N-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2\log N - \log 8$. So let

$$\Lambda_N = 2L - \log 8, \qquad G(x) = x + \log x,$$

Table 1: Elephant crossings x_N and the root g_N of $G(g) = \Lambda_N$. These values are numerical and are not used in any proof. We computed x_N in double precision from the recursion of Lemma 2.2, by Brent's method with an absolute tolerance of 10^{-12} in x, and we found g_N with a tolerance of about 10^{-15}.

N	x_N	g_N	$x_N/\log N$
50	3.940487	4.288636	1.007
400	7.629217	7.843768	1.273
1600	10.240565	10.340051	1.388
3200	11.546834	11.610464	1.431
12800	14.158845	14.182921	1.497
51200	16.778468	16.786946	1.547

and let g_N be the root of $G(g_N) = \Lambda_N$. It exists and is unique, since G increases from $-\infty$ to ∞ on $(0, \infty)$.

Theorem 2.7. *As $N \to \infty$ over even N, the following hold.*

(a) The crossing satisfies $x_N + \log x_N = \Lambda_N + O(L^2/N)$, so $x_N = g_N + O(L^2/N)$.

(b) The crossing satisfies $x_N = 2\log N - \log\log N - \log 16 + O(\log\log N/\log N)$.

Proof. (a) *Step 1. Both sides near the crossing.* Put $F(x) = \log C_N(x/N) - \log E_N(x/N)$ for $0 < x < N$, where $C_N(\varepsilon)$ and $E_N(\varepsilon)$ are the center and the end at flip probability ε. By Proposition 2.3, F is strictly increasing and x_N is its only zero. We show that

$$|F(x) - G(x) + \Lambda_N| \le \rho_N \quad (1 \le x \le 2\Lambda_N), \qquad \text{with } \rho_N = O(L^2/N). \tag{$*$}$$

Let $1 \le x \le 2\Lambda_N$ and $\varepsilon = x/N$. Then $\varepsilon \le 4L/N$, so for large N both parts of Theorem 2.6 apply, and together they say that $\log C_N = \log \frac{4\varepsilon}{N+2} + O(\varepsilon L)$. Here $\log \frac{4\varepsilon}{N+2} = \log x + \log 4 - 2L - \log(1 + 2/N)$, and $\log(1 + 2/N) \le 2/N \le 2xL/N$ because $x \ge 1$. For the end we use $\log(1 - y) = -y + O(y^2)$ for $0 \le y \le \frac{1}{2}$ in $\log E_N = -\log 2 + (N-1)\log(1 - \frac{x}{N})$. So

$$\log C_N = \log x + \log 4 - 2L + O\Big(\frac{xL}{N}\Big), \qquad \log E_N = -\log 2 - x + O\Big(\frac{x^2}{N}\Big).$$

We subtract. Since $\log 4 + \log 2 = \log 8$, the main terms give $x + \log x - 2L + \log 8 = G(x) - \Lambda_N$, and both errors are $O(L^2/N)$ because $x \le 2\Lambda_N \le 4L$. This is $(*)$.

Keeping the constants gives an explicit ρ_N. Suppose that $400L(1+L) \le N$. Then $\varepsilon(1+\log N) \le 4L(1+L)/N \le \frac{1}{100}$, so the inequality after part (b) of Theorem 2.6 applies. With $\log(1-y) \ge -y-y^2$ for $0 \le y \le \frac{1}{2}$ and $x \le 4L$, the error term for $\log C_N$ above lies between $-(3 + 8L)/N$ and $(8L^2 + 20800L)/N$, and the one for $\log E_N$ lies between $-16L^2/N$ and $4L/N$. So $(*)$ holds with $\rho_N = (24L^2 + 20800L)/N$.

Step 2. Trap the zero. We have $G' = 1 + 1/x \ge 1$ and $G(g_N) = \Lambda_N$. So $G(x) - \Lambda_N < -\rho_N$ for $x < g_N - \rho_N$, and $G(x) - \Lambda_N > \rho_N$ for $x > g_N + \rho_N$. By $(*)$, $F(x) < 0$ for $1 \le x < g_N - \rho_N$ and $F(x) > 0$ for $g_N + \rho_N < x \le 2\Lambda_N$. Both ranges are nonempty for large N, because $\Lambda_N - \log\Lambda_N \le g_N \le \Lambda_N$ (note that $G(\Lambda_N - \log\Lambda_N) \le \Lambda_N \le G(\Lambda_N)$). Since F is increasing, its zero x_N lies between the two ranges, so $|x_N - g_N| \le \rho_N$. Then x_N lies in $[1, 2\Lambda_N]$, and $F(x_N) = 0$ in $(*)$ gives $|G(x_N) - \Lambda_N| \le \rho_N$. With the explicit ρ_N above, these two bounds hold for every even N with $L \ge 30$. Indeed $e^L \ge L^{12}/12!$ gives $24L^4 + 20800L^3 \le N$, so $400L(1 + L) \le N$ and $\rho_N \le 1/L^2$, and both ranges are nonempty because $g_N \ge \Lambda_N - \log\Lambda_N > 50$.

(b) We solve $g_N = \Lambda_N - \log g_N$ by one substitution. Since $\Lambda_N - \log \Lambda_N \le g_N \le \Lambda_N$, the ratio g_N/Λ_N is $1 + O(\log \Lambda_N/\Lambda_N)$, so $\log g_N = \log \Lambda_N + O(\log \Lambda_N/\Lambda_N)$. Also $\Lambda_N = 2\log N - \log 8$ gives $\log \Lambda_N = \log 2 + \log\log N + O(1/\log N)$. So $g_N = 2\log N - \log 8 - \log 2 - \log\log N + O(\log\log N/\log N)$, which is (b) for g_N, and (a) gives it for x_N. □

Table 1 gives crossings for six values of N of the form $50 \cdot 2^k$. The limit 2 of $x_N/\log N$ is approached slowly, because of the term $-\log\log N$.

In Online Resource 1, Proposition S1.1 proves a lower bound for C_N that matches Theorem 2.6(b) up to a factor $1 + O(\varepsilon)$, and Proposition S1.2 gives one more term of the crossing, which shows that $x_N < g_N$ for large N. Theorem S1.7 there compares x_N with the crossing value z_N of T for the persistent walk. It shows that $x_N - (2z_N - \log(2\pi))$ is $c_*/\log N$ to first order, with $c_* = \frac{1}{2}\log(2\pi) - \frac{1}{4}$.

3 The shape of the law at the crossing

We now look at the whole law P_N near the crossing. In this section and in Appendices B and C, a statement that says "Assume (H)" assumes

$$N \ge 64, \qquad 0 < \varepsilon \le \tfrac{1}{4}, \qquad x = N\varepsilon \ge 1. \tag{H}$$

Every other statement holds under its own hypotheses. Recall that $P_N(N-m) = \frac{1}{2}[f_N(m) + f_N(N-m)]$. Some statements also use odd N. There x_N is the value of x at which the two central bins are as likely as the ends (it exists and is unique by Proposition 3.25). For a mode $a^* \ge N/2$ of P_N we put $m^* = N - a^*$, and m_h is the largest mode of f_N.

3.1 Families and the first flip

For $k_0 \ge 1$ let $(M_k^{[k_0]})_{k \ge k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M. It is the count of disagreeing steps for a walk whose first k_0 steps are forced to agree. So $M^{[1]}$ has the law of M. We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \ge 2$ and $t \ge j-1$. In a random recursive tree on $[N] = \{1, \dots, N\}$, vertex $k+1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, N-1$. We write $\mathcal{T}_v$ for the subtree of v, that is, v and its descendants, and we call it the family of v.

Lemma 3.1 (tree and marks)**.** *Let $\mathcal{T}$ be the tree on $[N]$ in which the parent of $k+1$ is U_k. For $2 \le v \le N$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then $\mathcal{T}$ is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of $\mathcal{T}$, and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_N = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \ge k_0$, is a copy of the chain $M^{[k_0]}$.*

Proof. The parent U_k of $k+1$ is uniform on $[k]$ and independent of the earlier choices, and the marks are the flip indicators. The path from $k+1$ to the root is $k+1$ followed by the path from U_k, so $\sigma(k+1) = \mu_{k+1} + \sigma(U_k)$ and, by induction on k, $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \cdots = \mu_{k_0} = 0$. If b of the vertices in $[k]$ are odd, $k \ge k_0$, then $k+1$ is odd with conditional probability $(1-\varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. □

So a single flip at v turns every step of its family (Figure 1). For $k \ge v$, if $\mathcal{T}_v$ contains s of the vertices in $[k]$, then $k+1$ joins $\mathcal{T}_v$ with probability s/k. This is the urn of Lemma 2.4, so $|\mathcal{T}_v|$

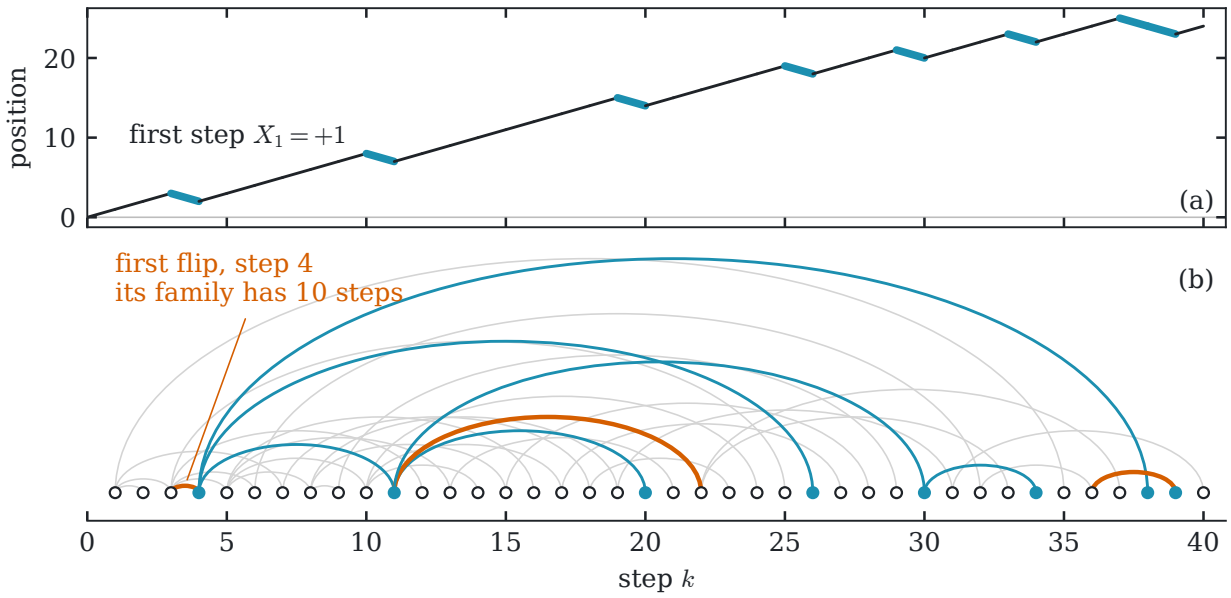


Figure 1: One elephant walk with $N = 40$ steps and $\varepsilon = 0.07$. (a) The position against the step. The steps that disagree with the first step are drawn heavier and in blue. (b) The memory tree of the same walk. Step $k+1$ copies the step at the left end of its arc, and the arcs drawn heavier are flips. A step disagrees with the first step exactly when the path from it to step 1 crosses an odd number of flips (Lemma 3.1). The first flip, at step 4, turns its family of ten steps. The flip at step 22, inside that family, turns its own family of three steps back, and the flip at step 39 turns only that step.

has the law β_v and mean N/v, and on average there are $N/(d(d+1))$ families of size d. If v is the only marked vertex, the odd vertices are those of $\mathcal{T}_v$. Two or more marks have probability $O(\varepsilon^2)$ at fixed N, and no mark gives $M_N = 0$. So at fixed N the part of $f_N(m)$ of first order in ε is exactly $\varepsilon \sum_v \beta_v(m) = x/(m(m+1))$ for $1 \le m \le N-1$, which is the formula of [19, Eq. (10)] with one cell at the start. It is also proved in Lean for the chain M (Section 5).

To go beyond first order we cut the tree at the first flip.

Lemma 3.2 (splitting)**.** *Let $\mathcal{T}$ be a random recursive tree on $[N]$, let $2 \le v \le N$ and let $S \subseteq [N]$ with $\min S = v$. Given that $\mathcal{T}_v$ has vertex set S, the restrictions of $\mathcal{T}$ to S and to $[N] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v, which is uniform on $[v-1]$.*

Proof. The tree $\mathcal{T}$ is uniform over the $(N-1)!$ increasing trees on $[N]$. Its family $\mathcal{T}_v$ has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \ge 2$ has its parent outside S. So, given this event, each element of S other than v picks its parent freely among the smaller elements of S, v picks any element of $[v-1]$, and each element of $[N] \setminus S$ other than 1 picks freely among the smaller elements of $[N] \setminus S$. This is the claim. □

Lemma 3.3 (the first flip)**.** *Let $V = \min\{v \ge 2 : \mu_v = 1\}$, with $V = \infty$ and $|\mathcal{T}_V| = 0$ if there is no mark. If $V < \infty$, let $\bar{Y}$ be the number of vertices u of $\mathcal{T}_V$ with $\sigma(u)$ even, and let $\bar{Z}$ be the number of odd vertices outside $\mathcal{T}_V$. Then for $2 \le j \le N$, $1 \le d \le N - j + 1$ and integers y, z,*

$$\mathbb{P}\big(V = j,\, |\mathcal{T}_V| = d,\, \bar{Y} = y,\, \bar{Z} = z\big) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)\,\mathbb{P}(Y_d = y)\,\mathbb{P}\big(Z^{(j)}_{N-d} = z\big).$$

In particular $\mathbb{P}(V = j,\, |\mathcal{T}_V| = d) = \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, *and for* $m \geq 1$

$$f_N(m) = \sum_{j=2}^{N} \varepsilon(1-\varepsilon)^{j-2} \sum_{d=1}^{N-j+1} \beta_j(d)\, \mathbb{P}\big(d - Y_d + Z_{N-d}^{(j)} = m\big),$$

where Y_d *and* $Z_{N-d}^{(j)}$ *are independent.*

Proof. Marks and tree are independent, and $|\mathcal{T}_j|$ has the law β_j, which gives the law of $(V, |\mathcal{T}_V|)$. Condition on $V = j$ and on the vertex set S of $\mathcal{T}_j$, with $|S| = d$. The marks on $j+1, \ldots, N$ are still i.i.d., and by Lemma 3.2 the trees on S and on $[N] \setminus S$ are independent random recursive trees. The vertices $1, \ldots, j-1$ carry no mark, so j is odd, and $u \in S$ is even exactly when the path from u up to j, with j excluded, has an odd number of marks. So by Lemma 3.1 for the tree on S rooted at j, the count $\bar{Y}$ has the law of Y_d. Paths from outside S stay outside S, and the tree on $[N] \setminus S$ has the unmarked vertices $1, \ldots, j-1$ first. So by Lemma 3.1 the count $\bar{Z}$ has the law of $Z_{N-d}^{(j)}$. The two counts are functions of the two trees and of the marks on the two disjoint sets S and $[N] \setminus S$, so they are independent, and their joint law depends on S only through d. This gives the joint law. On $\{V = j\}$ we have $M_N = d - \bar{Y} + \bar{Z}$, and $M_N = 0$ if there is no mark. Summing over j, d and the pairs (y, z) with $d - y + z = m$ gives the formula for $f_N(m)$. □

So the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j-1$ steps agree with X_1. Only the law of each pair $(Y_d, Z_{N-d}^{(j)})$ enters the lemma. So from now on $(Y_d)_{d \geq 1}$ is a copy of the chain $M^{[1]}$, and $(Z_t^{(j)})_{t \geq j-1}$ is a copy of $M^{[j-1]}$, independent of Y and of $(V, |\mathcal{T}_V|)$.

Lemma 3.4 (unimodality given the first step [7, Section 2, proof of Theorem 1.1])**.** *The function* f_N *is unimodal.*

This is shown at the start of the proof of Theorem 1.1 of [7], for every N and every memory parameter $p = 1 - \varepsilon \in [0, 1]$, for the position $N - 2M_N$ given $X_1 = +1$. The lemma is also proved in Lean (Section 5). So f_N is nondecreasing on $\{0, \ldots, m_h\}$ and nonincreasing on $\{m_h, \ldots, N\}$. With a fair first step, P_N mixes f_N and its mirror image. When the mode of f_N is far from the center, the mixture can have two humps, one near each end.

3.2 The end and its neighbor

We first say heuristically why the end should fall below its neighbor once $x > 2$ (Figure 2(a), inset). Given $X_1 = +1$, the end needs no flip at all. Its neighbor, bin $N-1$, needs one flip that is never copied. A flip at step j is never copied with probability about j/N. Summing $\varepsilon j/N$ over j gives a factor of about $x/2$ in favor of the neighbor.

Theorem 3.5 (endpoint dip)**.** *Let* $N \geq 2$ *and* $0 < \varepsilon < 1$, *and put* $\rho_1 = P_N(N-1)/E_N$. *Then*

$$\rho_1 > \frac{x}{2(1-\varepsilon)}.$$

If moreover $N \geq 3$ *and* $\varepsilon \leq \frac{1}{2}$, *then*

$$\rho_1 \leq \frac{x(1+1/N)^2}{2-3\varepsilon} + \frac{\varepsilon}{1-\varepsilon}.$$

Hence the end is strictly below its neighbor if $x \geq 2(1-\varepsilon)$. *If* $N \geq 3$, $\varepsilon \leq \frac{1}{2}$ *and the right side of the second bound is below* 1, *the end is strictly above its neighbor.*

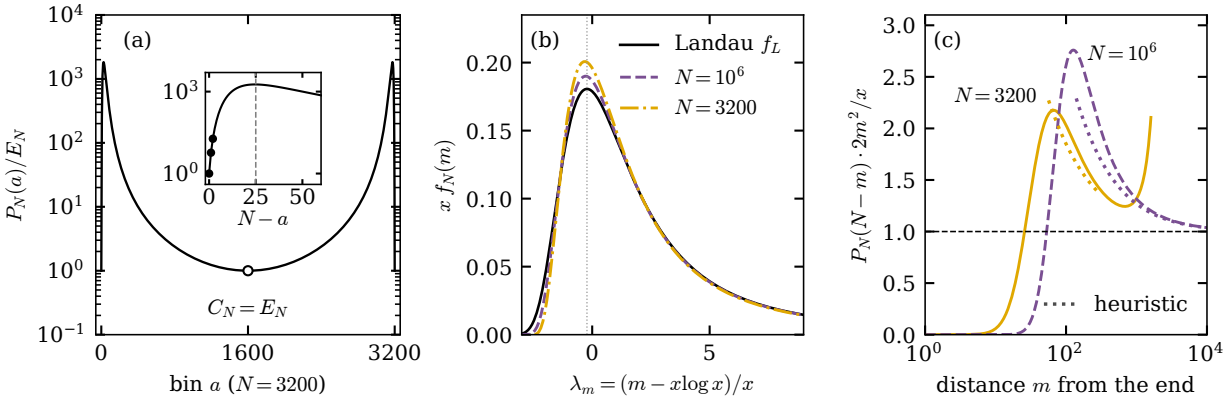


Figure 2: The elephant walk at its crossing. (a) The law P_N for $N = 3200$ at its crossing $x = 11.5468$, divided by the end E_N. The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins N, $N-1$ and $N-2$. The end is below its neighbor, and the modes are 25 bins from each end (dashed). (b) The law of M_N, scaled as $xf_N(m)$, against $\lambda_m = (m - x\log x)/x$, with the Landau density f_L. The dotted line is its mode λ_{L}. (c) The tail $P_N(N-m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem S2.1 of Online Resource 1, and the dotted curves are the heuristic correction $1 + x(2\log m + 2\gamma - 3)/m$ for $2x\log x \le m \le N/x$. The rise of the $N = 3200$ curve near $m = N/2$ is the mirror term $f_N(N-m)$. At $N = 10^6$ we use $x = g_N = 22.4407$.

Proof. By Lemma 2.2, $\rho_1 = \big(f_N(1) + f_N(N-1)\big)/f_N(0)$ with $f_N(0) = (1-\varepsilon)^{N-1}$. We bound the two terms in turn.

The term $f_N(1)$. On the event $M_N = 1$ the chain moves up exactly once, say from time $j-1$ to time j with $2 \le j \le N$. Before that it stays at 0 for $j-2$ steps, each with probability $1 - r_t(0) = 1 - \varepsilon$. The move up has probability ε. After it the chain stays at 1 at the times $k = j, \dots, N-1$, each with probability $1 - r_k(1) = (1-\varepsilon)(1 - c'/k)$, where $c' = (1-2\varepsilon)/(1-\varepsilon) \le 1$. Multiplying and summing over j gives

$$f_N(1) = \varepsilon(1-\varepsilon)^{N-2} \sum_{j=2}^{N} \vartheta_j, \qquad \vartheta_j = \prod_{k=j}^{N-1} \Big(1 - \frac{c'}{k}\Big).$$

For the lower bound we use $c' \le 1$. Then $\vartheta_j \ge \prod_{k=j}^{N-1}(1 - 1/k) = (j-1)/(N-1)$, and these lower bounds sum to $N/2$ over $2 \le j \le N$. So $f_N(1)/f_N(0) \ge x/(2(1-\varepsilon))$. For the upper bound let $N \ge 3$ and $\varepsilon \le \frac{1}{2}$, so that $c' \ge 0$. Then $1 - c'/k \le e^{-c'/k}$ and $\sum_{k=j}^{N-1} 1/k \ge \log(N/j)$ give $\vartheta_j \le (j/N)^{c'}$. As $(t/N)^{c'}$ is nondecreasing in t, we can compare the sum with an integral, namely

$$\sum_{j=2}^{N} \vartheta_j \le \int_2^{N+1} \Big(\frac{t}{N}\Big)^{c'} dt \le \frac{N(1+1/N)^{c'+1}}{c'+1} \le \frac{N(1+1/N)^2(1-\varepsilon)}{2-3\varepsilon},$$

using $c' \le 1$ and $1/(c'+1) = (1-\varepsilon)/(2-3\varepsilon)$. So $f_N(1)/f_N(0) \le x(1+1/N)^2/(2-3\varepsilon)$.

The term $f_N(N-1)$. The event $M_N = N-1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \ge 2$ has probability $r_k(k-1) = 1 - \varepsilon - (1-2\varepsilon)/k$. This lies strictly between 0 and 1, as every $r_k(b)$ with $0 < b < k$ does, and it lies in $[\frac{1}{2}, 1-\varepsilon]$ when $\varepsilon \le \frac{1}{2}$. So $f_N(N-1) > 0$, and $f_N(N-1) \le \varepsilon(1-\varepsilon)^{N-2}$ when $\varepsilon \le \frac{1}{2}$. This makes the lower bound

strict and gives the term $\varepsilon/(1-\varepsilon)$ in the upper bound. (For $N=2$ the bins 1 and $N-1$ coincide, and the argument is the same.) □

Corollary 3.6 (the dip at the crossing)**.** *Let $N \geq 2$ be even and $0 < \varepsilon < 1$, and suppose that $x \geq 2(1-\varepsilon)$ and $C_N \leq E_N$. Then $C_N \leq E_N < P_N(N-1)$, and P_N is not unimodal. In particular, if $x_N \geq 2(1-\varepsilon_N)$, then at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal.*

Proof. Theorem 3.5 gives $P_N(N-1) > E_N \geq C_N$. Suppose that P_N is nondecreasing on $\{0,\dots,a_0\}$ and nonincreasing on $\{a_0,\dots,N\}$. It is symmetric about h, so we may replace a_0 by $N-a_0$ and assume $a_0 \leq h$. Then $P_N(a) \leq C_N$ for every $a \geq h$, including $a = N-1$, which is a contradiction. □

By Proposition 2.3, $C_N \leq E_N$ holds exactly when $\varepsilon \leq \varepsilon_N$, and $x \geq 2(1-\varepsilon)$ says that $\varepsilon \geq 2/(N+2)$. So the corollary covers the ε with $2/(N+2) \leq \varepsilon \leq \varepsilon_N$. In particular the condition $x_N \geq 2(1-\varepsilon_N)$ says that $\varepsilon_N \geq 2/(N+2)$. By Proposition 2.3 it holds exactly when $C_N \leq E_N$ at $\varepsilon = 2/(N+2)$. We now show that this is so for every even $N \geq 8$.

Proposition 3.7 (the hypothesis holds from $N=8$ on, computer-assisted)**.** *For every even $N \geq 8$ we have $C_N < E_N$ at $\varepsilon = 2/(N+2)$, and hence $x_N > 2(1-\varepsilon_N)$. So for every even $N \geq 8$, at the crossing $E_N = C_N < P_N(N-1)$, and P_N is not unimodal. For $N = 2$, 4 and 6 the condition $x_N \geq 2(1-\varepsilon_N)$ fails, and no ε meets the hypotheses of Corollary 3.6.*

Proof. The range $N \geq 1688$. Take $\varepsilon = 2/N$. Then $\varepsilon(1+\log N) \leq \frac{1}{100}$, so Theorem 2.6(b) gives $C_N \leq \frac{8}{N(N+2)} N^{4/N} e^{10400/N}$. The right side decreases in N and is below $1.4 \cdot 10^{-3}$ at $N = 1688$. Since $\log(1-y) \geq -y - y^2$ for $0 \leq y \leq \frac{1}{2}$, we also have $E_N = \frac{1}{2}(1-\frac{2}{N})^{N-1} \geq \frac{1}{2} e^{-2-2/N} > 0.06$. So $C_N < E_N$ at $\varepsilon = 2/N$, and Proposition 2.3 gives $\varepsilon_N > 2/N$. Hence $C_N < E_N$ at $\varepsilon = 2/(N+2)$ as well.

The range $8 \leq N < 1688$. For $N = 8$ the ratio C_N/E_N at $\varepsilon = 2/(N+2)$ is exactly $\frac{67903}{71680} < 1$. For even $10 \leq N \leq 2000$ the script `certify_center.py` encloses ε_N in ball arithmetic as in the proof of Proposition 3.24, and it checks by an exact rational comparison that the lower end of the enclosure lies above $2/(N+2)$. So $\varepsilon_N > 2/(N+2)$, and Proposition 2.3 gives $C_N < E_N$ at $\varepsilon = 2/(N+2)$. The ratio of the lower end to $2/(N+2)$ is at least 1.107, with its minimum at $N = 10$.

The three smallest N. At $\varepsilon = 2/(N+2)$ the ratio C_N/E_N is exactly 2, $\frac{5}{3}$ and $\frac{508}{405}$ for $N = 2$, 4 and 6, so $\varepsilon_N < 2/(N+2)$ there. These ratios and the one for $N = 8$ are computed in exact rational arithmetic by `certify_elephant.py`. The last claim of the proposition about $N \geq 8$ is Corollary 3.6. □

3.3 The bulk

The steps that disagree with X_1 lie in the flipped families. Recall the compound Poisson law CP_N of Section 1.2. On average $x/(d(d+1))$ flipped vertices have a family of size d. If no flipped step has a flipped descendant, which is likely when $x^2 \log N/N$ is small, then M_N is the total size of the flipped families, and this is close to the compound Poisson sum.

The CP law and its Landau limit are known (Section 1.3). We give an explicit total variation bound for the CP approximation to the elephant law, and an explicit local rate for the law with jump sizes truncated at $N-1$. Numerical values of f_L and its mode are standard (see Kölbig and Schorr [26]). Lemma 3.10 adds a certified enclosure.

Theorem 3.8 (compound Poisson bulk). *For $N \geq 1$ and $0 < \varepsilon < 1$,*

$$\mathrm{TV}(f_N, \mathrm{CP}_N) \leq \delta_\Sigma := \varepsilon^2(NH_N - 2N + 1) + 2\varepsilon^2(N-1) + 4\varepsilon^2 \sum_{k=0}^{N-2} (H_k^2 + H_k).$$

Also $\delta_\Sigma \leq \delta_B := \frac{x^2 \log N}{N} + \frac{4x^2}{N} e^{4x^2/N} + \frac{4x^2(H_N^2 + H_N)}{N}$.

Here $\mathrm{TV}(P, Q) = \frac{1}{2}\sum_k |P(k) - Q(k)|$. Thus the same bound holds for each mass and each event. Also $\delta_\Sigma = O(x^2 \log^2 N/N)$. Appendix C.1 proves the bound by coupling marks with Poisson variables and revealing the tree's parents one at a time. Changing one parent affects family sizes only along two paths to the root, whose depths have second moments of order $\log^2 N$.

The families of size up to about x add up to nearly $x \log x$, and the few families of size of order x give fluctuations of order x. So we center M_N at $x \log x$ and divide by x, which gives the coordinate λ_m and the Landau density f_L of Section 1.2. It is a stable density with index one. We show in Appendix C that e^B is the characteristic function of a probability law of class L, so that f_L is a unimodal probability density by [27].

Theorem 3.9 (Landau limit). *(a) Let $x \geq 1$ and $N \geq 4x$. For every integer m, with $\mathrm{CP}_N(m) = 0$ for $m < 0$,*

$$\big|x\,\mathrm{CP}_N(m) - f_L(\lambda_m)\big| \leq E_0 := \frac{889 + 165\log x}{x} + \frac{6.6\,x}{N} \leq 889\Big(\frac{1 + \log x}{x} + \frac{x}{N}\Big).$$

(b) Let $0 < \varepsilon \leq \frac{1}{4}$ and $x \geq 1$. For every integer $m \geq 0$,

$$\big|x f_N(m) - f_L(\lambda_m)\big| \leq E_0 + x\delta_\Sigma \leq 889\Big(\frac{1 + \log x}{x} + \frac{x}{N}\Big) + x\delta_B.$$

(c) Let $N_n \geq 1$ and $0 < \varepsilon_n < 1$ for $n \geq 1$, and put $x_n = N_n\varepsilon_n$. If $x_n \to \infty$ and $x_n^3 \log^2 N_n/N_n \to 0$, then $(M_{N_n} - x_n \log x_n)/x_n$ converges in law to the law with density f_L.

Near the crossing the number of disagreeing steps is about $x \log x$, with fluctuations of order x that follow the Landau law, bin by bin (Figure 2(b)). Both conditions of (c) hold at the crossing, by Theorem 2.7.

Appendix C proves the local bound by expanding the compound Poisson exponent at zero and applying Fourier inversion. The expansion gives the centering $x \log x$, and its remainder gives the rate.

To locate the modes we need to know that f_L is curved at its maximum.

Lemma 3.10 (the mode of the Landau density, computer-assisted). *(a) The density f_L has exactly one mode λ_{L}, and $-0.22278299 < \lambda_{\mathrm{L}} < -0.22278297$. It is nondecreasing on $(-\infty, \lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}}, \infty)$.*

(b) The derivative vanishes at the mode, $f_L'(\lambda_{\mathrm{L}}) = 0$, and $-0.0792 \leq f_L''(\lambda) \leq -0.0786$ for $-0.2229 \leq \lambda \leq -0.2227$.

(c) For every real λ, $f_L(\lambda_{\mathrm{L}}) - f_L(\lambda) \geq \kappa_L \min\{1, (\lambda - \lambda_{\mathrm{L}})^2\}$ with $\kappa_L = 2.6 \cdot 10^{-10}$.

The proof is in Appendix C.3, and it is computer-assisted. Lean proves the lemma as well, with the unimodality of f_L that [27] gives as its only input (Section 5), so the computation in ball arithmetic is an independent check.

Corollary 3.11 (where the modes are). *(a) Assume* (H). *Let κ_L be as in Lemma 3.10(c), let $E_1 = 889\big(\frac{1+\log x}{x} + \frac{x}{N}\big) + x\delta_B$ be the bound of Theorem 3.9(b), and put*

$$E' = E_1 + \frac{16x^2 H_N}{N^2} + \frac{\|f_L''\|_\infty}{8x^2}.$$

If $E' < \kappa_L/2$ and $m_h \le N/4$, then every mode $a^* \ge N/2$ of P_N satisfies $|m^* - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\kappa_L}$, and so does every mode of f_N.

(b) There are K and N_2 such that the following holds for every $N \ge N_2$, even or odd, and every ε with $\log N \le x \le 2\log N$. Every mode $a^* \ge N/2$ of P_N has

$$\big|m^* - x\log x - \lambda_{\mathrm{L}} x\big| \le K\sqrt{x\log x},$$

and so has every mode of f_N in place of m^*. In particular this holds at the crossing, with $x = x_N$, for every large even N (Theorem 2.7(b)).

In (a) every constant is explicit. (Here $\|f_L''\|_\infty \le 16/\pi^4$, since $|e^{B(t)}| = e^{-\pi|t|/2}$.) We do not compute K and N_2 in (b).

So each hump of the law at the crossing sits about $x\log x - 0.2228\,x$ bins from its end. The proof is in Appendix C.4. It compares f_N at a mode with f_N at the integer nearest to $x\log x + \lambda_{\mathrm{L}} x$, and uses Theorem 3.9(b) and Lemma 3.10(c).

Remark 3.12 (how good the bulk approximation is). The bounds of this subsection are far from sharp at the sizes we can compute. At $N = 10^6$ and $x = 22.4407$ the bound δ_B is still about 0.46, but there the computed law has $|f_N(m)/\mathrm{CP}_N(m) - 1| \le 3.6 \cdot 10^{-4}$ for $1 \le m \le 10^4$. At every tested crossing m^* equals the mode of CP_N. The proof of Corollary 3.11(b) needs the bound of Theorem 3.9(b) to be below $\kappa_L/2$, where $\kappa_L = 2.6 \cdot 10^{-10}$ is the constant of Lemma 3.10(c). With the constant 889 this needs $x > 2 \cdot 10^{14}$, and numerically the best κ_L is about 0.026. These numbers are computed in double precision and are not used in any proof.

3.4 The tail

Beyond the humps the law falls like $x/(2m^2)$. Theorem S2.1 of Online Resource 1 proves that $P_N(N-m) = \frac{x}{2m^2}(1+o(1))$ for even N with $x \ge 1$ and $x\log^2 N/N \to 0$, uniformly for $\omega_N x\log^2(x+1) \le m \le \delta'_N N$, for every $\omega_N \to \infty$ and every $\delta'_N \to 0$.

At $N = 10^6$, $x = 22.4407$ and $m = 10^4$ the computed ratio $P_N(N-m)/(x/(2m^2))$ is 1.03777. Heuristically, at moderate m the law CP_N suggests $f_N(m)\,m(m+1)/x \approx 1 + x(2\log m + 2\gamma - 3)/m$, which we do not prove. This factor is still about 2 near $m = 70$ when $N = 3200$ (Figure 2(c)).

3.5 The shape at every even crossing

This subsection proves that the second difference of P_N changes sign at most four times. At every even crossing with $N \ge 8$ this gives the unique interior minimum and the two intervals of modes, with no asymptotic input.

Throughout this subsection and the next, $0 < \varepsilon < \frac{1}{2}$ and $c = 1 - 2\varepsilon$, so that $r_t(b) = \varepsilon + cb/t$. For $t \ge 1$ and every integer a we write $P_t(a) = \mathbb{P}(A_t = a)$, which is 0 unless $0 \le a \le t$, and

$$D_t(a) = P_t(a+1) - 2P_t(a) + P_t(a-1).$$

We also put $s_t = D_t(1) = P_t(0) - 2P_t(1) + P_t(2)$ for $t \ge 2$.

Lemma 3.13 (recurrence for the second difference)**.** *For every $t \geq 1$ and every integer a,*

$$P_{t+1}(a) = \big(1 - r_t(a)\big) P_t(a) + r_t(a-1)\, P_t(a-1), \tag{1}$$
$$D_{t+1}(a) = \big(1 - r_t(a+1)\big) D_t(a) + r_t(a-2)\, D_t(a-1). \tag{2}$$

Also $P_t(a) = P_t(t-a)$ and $D_t(a) = D_t(t-a)$. The same two recurrences hold with $f_t(m)$ in place of $P_t(a)$.

Proof. Given $A_t = a$, the copied step is $+1$ with probability a/t, so step $t+1$ is $+1$ with probability $(1-\varepsilon)\frac{a}{t} + \varepsilon(1 - \frac{a}{t}) = r_t(a)$. This gives (1) for $0 \leq a \leq t+1$, where at $a = 0$ and at $a = t+1$ the formula has one extra term, namely $r_t(-1)P_t(-1)$ or $(1 - r_t(t+1))P_t(t+1)$, and this term is 0 because $P_t(-1) = P_t(t+1) = 0$. For other a every term vanishes. The symmetry of P_t is Lemma 2.2, and it gives the symmetry of D_t. The chain M of Lemma 2.2 has the same transition probabilities, so (1) holds for f_t as well.

For (2) put $\varsigma = c/t$, so that $r_t(b+1) = r_t(b) + \varsigma$. We insert (1) at $a+1$, a and $a-1$ into the definition of $D_{t+1}(a)$ and collect the coefficients of $P_t(a+1)$, $P_t(a)$, $P_t(a-1)$ and $P_t(a-2)$. They are

$$1 - r_t(a+1), \qquad 3r_t(a) - 2, \qquad 1 - 3r_t(a-1), \qquad r_t(a-2).$$

The right side of (2) has the coefficients $1 - r_t(a+1)$, $-2 + 2r_t(a+1) + r_t(a-2)$, $1 - r_t(a+1) - 2r_t(a-2)$ and $r_t(a-2)$. The second ones agree because $2r_t(a+1) + r_t(a-2) = 3r_t(a)$, and the third ones agree because $r_t(a+1) + 2r_t(a-2) = 3r_t(a-1)$. □

Trinas and Valle [8, (2.2)] write (1) for the mass function of the position, which is the forward equation of the counting chain of Lemma 2.2, and they use it for a local limit theorem in the superdiffusive regime. In [8, (4.10)] they write the analogue of (2) for differences of higher order, and they use it at a fixed memory parameter in a certified study of log-concavity. We use (2) for signs. The coefficients in (2) are positive when $2 \leq a \leq t-1$. So the second difference moves like a mass that is partly killed, and its sign can change only through what enters at the two edges. The next lemma is the elementary form of the variation diminishing property that we need.

For a finite list z of real numbers let $\mathrm{sp}(z)$ be its sign pattern. We delete the zeros of z, replace each entry by its sign, and merge equal neighbors. So $\mathrm{sp}(z)$ is a list of alternating signs. For two such lists we write $p \preceq q$ if p is a subsequence of q. We put $\Pi_5 = (+,-,+,-,+)$ and $\Pi_3 = (+,-,+)$.

Lemma 3.14 (one step)**.** *Let $\zeta = (\zeta_1, \dots, \zeta_n)$ be real, and let $\eta_i = A_i\zeta_i + B_i\zeta_{i+1}$ for $1 \leq i \leq n-1$ with $A_i \geq 0$ and $B_i \geq 0$. Let $w = (\zeta_1, \eta_1, \zeta_2, \eta_2, \dots, \eta_{n-1}, \zeta_n)$. Then $\mathrm{sp}(w) = \mathrm{sp}(\zeta)$ and $\mathrm{sp}(\eta) \preceq \mathrm{sp}(\zeta)$.*

We prove this in Appendix E.1.

Lemma 3.15 (the sequence next to the end)**.** *For every $t_1 \geq 2$ we have $\mathrm{sp}(s_2, s_3, \dots, s_{t_1}) \preceq \Pi_3$. Also, for every $0 \leq \varepsilon \leq 1$, the numbers s_3 and s_4 do not both vanish.*

We prove this in Appendix E.2.

Theorem 3.16 (at most five blocks)**.** *Let $N \geq 3$ and $0 < \varepsilon < \frac{1}{2}$. Then*

$$\mathrm{sp}\big(D_N(1), D_N(2), \dots, D_N(N-1)\big) \preceq (+,-,+,-,+).$$

So the second difference of P_N changes sign at most four times on $\{1, \dots, N-1\}$. The same holds for the second difference of f_N, and for f_N it holds for $N = 2$ as well.

We prove this in Appendix E.3.

The next theorem is the step that turns one sign into a whole convex stretch. Its idea is short. A sign change from negative to positive at $a_0 < a_1$, together with its mirror image, gives the signs $-, +, +, -$ at $a_0 < a_1 < N - a_1 < N - a_0$. A negative second difference in between would add a fifth alternating sign, which Theorem 3.16 forbids. A zero in between is subtler. We follow it back in time through (2), whose coefficients are positive, so a zero has two zero parents unless its parents have opposite signs, and opposite signs would again give five alternating signs. Going back to $t = 5$ forces $s_3 = s_4 = 0$, which Lemma 3.15 rules out.

Theorem 3.17 (no holes)**.** *Let $N \geq 5$ and $0 < \varepsilon < \frac{1}{2}$. Let $1 \leq a_0 < a_1 \leq N/2$ be integers with $D_N(a_0) < 0 < D_N(a_1)$. Then $D_N(a) > 0$ for every $a_1 \leq a \leq N - a_1$.*

We prove this in Appendix E.4.

Lemma 3.18 (the center from no holes)**.** *Let $N \geq 6$ be even and $0 < \varepsilon < \frac{1}{2}$. If $P_N(1) > \max\{E_N, C_N\}$, then $P_N(a) > C_N$ for $1 \leq a \leq N - 1$, $a \neq h$, and $D_N(h) > 0$.*

Proof. Suppose that $P_N(a) \leq C_N$ for some $1 \leq a < h$, and let k be the first minimizer of P_N on $\{1, \ldots, h\}$. Then $2 \leq k < h$ and $P_N(k-1) > P_N(k) \leq P_N(k+1)$, so $D_N(k) > 0$. The first maximizer a_0 on $\{0, \ldots, k\}$ has value at least $P_N(1)$, which exceeds both endpoint values of this interval. Thus $1 \leq a_0 < k$ and $D_N(a_0) < 0$. Theorem 3.17 gives $D_N(a) > 0$ for $k \leq a \leq N - k$.

Put $\delta(a) = P_N(a+1) - P_N(a)$. These first differences are strictly increasing on $\{k-1, \ldots, N-k\}$, and symmetry gives $\delta(h) = -\delta(h-1)$. Hence $\delta(h-1) < 0$, and every difference from k to $h-1$ is negative. This makes $P_N(k) > P_N(h)$, a contradiction. Reflection proves the other half, and $D_N(h) = 2(P_N(h-1) - P_N(h)) > 0$. □

Corollary 3.19 (the shape at every even crossing)**.** *For every even $N \geq 8$, at the crossing,*

$$P_N(a) > C_N = E_N \quad (1 \leq a \leq N-1,\ a \neq h), \qquad D_N(h) > 0.$$

There are integers $1 \leq l \leq r < h$ such that P_N strictly increases from 0 to l, is constant from l to r, and strictly decreases from r to h. Its two intervals of global maximizers are $\{l, \ldots, r\}$ and $\{N - r, \ldots, N - l\}$.

Proof. At the crossing $P_N(1) > E_N = C_N$ by Proposition 3.7, so Lemma 3.18 gives the minimum and central curvature. For the humps, let $p = \mathrm{sp}(D_N(1), \ldots, D_N(h))$. It ends in $+$, and symmetry makes the full pattern p followed by its reverse, with the central positive blocks merged. If p has j blocks, the full pattern has $2j - 1 \leq 5$ by Theorem 3.16. Thus p is a subsequence of $(+, -, +)$.

The first differences $\delta(a) = P_N(a+1) - P_N(a)$ on $0 \leq a < h$ therefore have at most three successive monotonic stretches, namely nondecreasing, nonincreasing, nondecreasing. Their first value is positive, and their last is $-D_N(h)/2 < 0$. The first stretch stays positive and the last stays negative. The middle stretch passes through zero at most once, possibly with consecutive zeros. This gives the claimed rise, plateau and fall. Symmetry gives the other hump. □

Corollary 3.20 (a window from the maximum)**.** *Let $N \geq 8$ be even and $x = x_N$. Let $[l, r]$ be the left interval of maximizers in Corollary 3.19, and put*

$$M = P_N(r), \qquad B = r - 1 + M^{-1}, \qquad b = \min\{h, \lceil B \rceil - 1\}.$$

Then $D_N(a) > 0$ for every integer $b \leq a \leq N - b$. This window is nonempty.

Proof. Let q be the first integer after l with $D_N(q) > 0$. Corollary 3.19 gives $D_N(l) < 0 < D_N(h)$, so $q \le h$. The constant plateau gives $q > r$. By its definition P_N is concave on $[r, q]$. Hence

$$\frac{M(q-r+1)}{2} \le \sum_{a=r}^{q-1} P_N(a) \le \frac{1-P_N(h)}{2} < \frac{1}{2},$$

where the first inequality sums the chord from $P_N(r) = M$ to the nonnegative value $P_N(q)$. Thus $q < B$, and consequently $q \le b \le h$. Theorem 3.17, applied at $l < q$, gives positivity on $[q, N-q]$, which contains $[b, N-b]$. □

The finite endpoint-dip certificates in Proposition 3.7 remain inputs. Neither corollary needs a mode-location or convex-window estimate. The height bound gives a window at every even length, although it may consist only of the center. Heuristically the window is much wider. Near the crossing, Theorem 3.9(b) and Corollary 3.11 give $M = P_N(r) = \frac{f_L(\lambda_L)}{2x}(1 + o(1))$, so with $m^* = r$ we get $B = m^* + \big(2/f_L(\lambda_L) + o(1)\big)x$. Since $f_L(\lambda_L) \approx 0.1807$, this is about $m^* + 11x$, far below h for large N. At $N = 4000$ the computed $B - m^*$ is about $9.9x$. We do not compute the $o(1)$, and the constants of Theorem 3.9 are far from sharp, so this is not a bound. The stronger prescribed window of Definition 3.21 is the subject of Section 3.6.

3.6 The prescribed window

This subsection concerns the stronger, prescribed convexity window of Definition 3.21. We prove it for large N and for $5 \le N \le 2000$, and it is open in between.

Definition 3.21 (convex between the humps)**.** Let $N \ge 5$. The law at the crossing is *convex between the humps* if, for $x = x_N$ and every mode $a^* \ge N/2$ of P_N with $m^* = N - a^*$, the second difference $P_N(a-1) - 2P_N(a) + P_N(a+1)$ is positive for $2m^* + 1 \le a \le N - 2m^* - 1$.

With the symmetry of the law, this makes P_N strictly convex on $[2m^*, N - 2m^*]$, so h is its unique minimizer there (the two central bins for odd N).

Corollary 3.22 (one inequality is enough)**.** *Let $N \ge 8$ be even and $x = x_N$, or let $N \ge 5$ and $0 < \varepsilon < \frac{1}{2}$ be such that $P_N(0) < P_N(1)$. Let a^* be any mode of P_N and $m^* = N - a^*$. If $D_N(a_1) > 0$ for some integer a_1 with $m^* < a_1 \le 2m^* + 1$ and $a_1 \le N/2$, then $D_N(a) > 0$ for every $2m^* + 1 \le a \le N - 2m^* - 1$. In particular P_N is strictly convex on $[2m^*, N - 2m^*]$ as soon as $D_N(2m^* + 1) > 0$.*

Proof. At the crossing with $N \ge 8$ even, $P_N(0) = E_N < P_N(N-1) = P_N(1)$ by Proposition 3.7. Let a_0 be the smallest maximizer of P_N. Then $1 \le a_0 \le m^*$, because $P_N(0) < P_N(1)$ and $P_N(m^*) = P_N(a^*)$ is the maximum. Also $P_N(a_0 - 1) < P_N(a_0) \ge P_N(a_0 + 1)$, so $D_N(a_0) < 0$. Now Theorem 3.17 applies to $a_0 < a_1$. □

The condition $a_1 \le N/2$ costs nothing at $a_1 = 2m^* + 1$, because $2m^* + 1 \le N/2$ whenever the range $2m^* + 1 \le a \le N - 2m^* - 1$ is not empty. If $a^* < N/2$, then $m^* > N/2$, and no a_1 meets the conditions.

Theorem 3.23 (convexity between the humps for large N)**.** *There is N_1 such that the following holds for every $N \ge N_1$, even or odd, and every ε with $\log N \le x \le 2 \log N$. For every mode $a^* \ge N/2$ of P_N with $m^* = N - a^*$ we have $4m^* + 4 \le N$ and*

$$P_N(a-1) - 2P_N(a) + P_N(a+1) > 0 \qquad \text{for } 2m^* + 1 \le a \le N - 2m^* - 1.$$

In particular, after enlarging N_1, the law at the crossing is convex between the humps for every even $N \ge N_1$. We do not compute the threshold N_1 (Remark 3.27).

Idea of the proof. If there were no point of strict convexity between m^* and $2m^*+1$, concavity would force more than half the total mass into an interval strictly to the left of the center. Symmetry rules this out, and Corollary 3.22 then gives the prescribed window.

Proof. Let $\log N \le x \le 2\log N$ and let $a^* \ge N/2$ be any mode. Write $m = m^* = N - a^*$ and $M = P_N(m)$. All estimates below are uniform in x and in the choice of mode, for either parity of N. By Corollary 3.11(b),

$$m = x\log x + \lambda_{\mathrm{L}} x + O(\sqrt{x\log x}), \qquad \lambda_m \longrightarrow \lambda_{\mathrm{L}}.$$

The error in Theorem 3.9(b) tends uniformly to zero, since $(1+\log x)/x = o(1)$ and $x\delta_B = O(\log^5 N/N)$. Consequently

$$2xM \ge x f_N(m) = f_L(\lambda_m) + o(1) = f_L(\lambda_{\mathrm{L}}) + o(1).$$

It follows that $M(m+3) > 1$, $m \ge 1$ and $2m+2 \le N/2$ for all sufficiently large N.

Suppose that $D_N(a) \le 0$ for every integer $m < a \le 2m+1$. Then P_N is concave on $[m, 2m+2]$, and $P_N(2m+2) \ge 0$ gives

$$\sum_{a=m}^{2m+1} P_N(a) \ge \frac{M}{m+2} \sum_{j=1}^{m+2} j = \frac{M(m+3)}{2} > \frac{1}{2}.$$

But this interval and its reflection are disjoint and have equal mass, a contradiction. Thus $D_N(a_1) > 0$ for some $m < a_1 \le 2m+1 \le N/2$. Theorem 3.5 gives $P_N(0) < P_N(1)$ because $x \ge \log N > 2$, and $\varepsilon < 1/2$ for large N. Corollary 3.22 now proves the required positivity. The final assertion follows from Theorem 2.7(b). □

Proposition 3.24 (small N, computer-assisted)**.** *For every $5 \le N \le N_{\mathrm{c}} = 2000$, even or odd, the law at the crossing is convex between the humps.*

Proof. For $10 \le N \le 2000$, the certified computation described in Section 5 brackets ε_N and encloses P_N throughout that bracket. It lists every possible mode and checks $D_N(a) > 0$ on each required range. When the range is nonempty, it also checks $D_N(a_0) < 0$ at some $1 \le a_0 \le 2m^*$ and $D_N(2m^*+1) > 0$. Theorem 3.17 then gives a second check of the conclusion.

For $5 \le N \le 9$, Lean brackets the crossing by exact rational arithmetic and Proposition 2.3. Over the whole bracket it checks $\varepsilon < \frac{1}{2}$ and $P_N(0) < P_N(1)$. For each possible mode it checks either that the prescribed window is empty, or that $2m^*+1 \le N/2$ and $D_N(2m^*+1) > 0$. Corollary 3.22 then gives the claim. These are outward-rounded integer computations with kernel-checked soundness (Section 5). At $N = 5$ the only possible right-hand mode is $a^* = 4$, whose window is empty. □

Proposition 3.25 (odd N)**.** *Let $N \ge 3$ be odd, and write $\widetilde{C}_N = P_N(\frac{N-1}{2}) = P_N(\frac{N+1}{2})$ for the two central bins. The ratio $\widetilde{C}_N/E_N$ is strictly increasing in ε, and it equals 1 at exactly one $\varepsilon_N \in (0, \frac{1}{2})$. Put $x_N = N\varepsilon_N$.*

(a) For every $0 < \varepsilon < \frac{1}{2}$,

$$\widetilde{C}_N = \Big(1 - \frac{c}{N}\Big)^{-1} C_{N+1},$$

where C_{N+1} is the center of the walk with $N+1$ steps and the same ε.

The center of a walk with an even number of steps is reached from the two central bins one step earlier, and from each of them with the same probability. So the two central bins of an odd N are the center of $N+1$ up to the factor $1-c/N$, and what we know about the crossing for even N carries over. Proposition S3.1 of Online Resource 1 shows that the crossing for odd N satisfies Theorem 2.7, that $P_N(0) < P_N(1)$ at the crossing for odd $N \geq 1689$, and that the law at the crossing is convex between the humps for every large odd N.

Proof. The crossing. By Proposition 2.3, $P_N(k)/E_N$ is strictly increasing in ε for $k = \frac{N+1}{2}$. It tends to 0 as $\varepsilon \to 0$. At $\varepsilon = \frac{1}{2}$ the steps are independent and fair, so $\widetilde{C}_N = \binom{N}{k} 2^{-N} > 2^{-N} = E_N$. So the ratio equals 1 at exactly one ε_N, and $\varepsilon_N < \frac{1}{2}$.

(a) Put $k = \frac{N+1}{2}$. Since $\varepsilon + \frac{c}{2} = \frac{1}{2}$, we have $r_N(k) = \frac{1}{2} + \frac{c}{2N}$ and $r_N(k-1) = \frac{1}{2} - \frac{c}{2N}$. So (1) with $t = N$ and $a = k$ gives

$$C_{N+1} = P_{N+1}(k) = \Big(\frac{1}{2} - \frac{c}{2N}\Big)\big(P_N(k) + P_N(k-1)\big) = \Big(1 - \frac{c}{N}\Big)\widetilde{C}_N.$$

□

In double precision the law at the crossing is also convex between the humps for $N = 3200$, 6400, 12800 and 25600. The definition leaves out the two endpoints $a = 2m^*$ and $a = N - 2m^*$, because there the second difference is not positive for some $N \leq 51$. Heuristically, between the humps $P_N(a)$ is close to $\frac{x}{2}\big[\frac{1}{a(a+1)} + \frac{1}{(N-a)(N-a+1)}\big]$, which is strictly convex in a. We also leave open the tail for m between about $x \log x$ and $x \log^2 x$, below the range of Theorem S2.1 of Online Resource 1.

The constants of the large-N result above are not computed, so the following is open.

Conjecture 3.26 (convexity between the humps)**.** *For every $N > 2000$ the law at the crossing is convex between the humps.*

Remark 3.27 (what is still open)*.* The remaining gap concerns the prescribed convexity window. The minimum and two-hump shape hold for every even $N \geq 8$, and Corollary 3.20 gives a nonempty convexity window there. The threshold N_1 in Theorem 3.23 is not computed, and its proof uses mode location and the local mass bound to obtain $P_N(m^*)(m^*+3) > 1$. At the crossing $m^* \approx x \log x$ and $P_N(m^*) \approx f_L(\lambda_L)/(2x)$, so this condition needs roughly $f_L(\lambda_L) \log x > 2$, that is x larger than about $6.4 \cdot 10^4$. Since $x_N \approx 2 \log N$, no computation can reach this. In double precision the product $P_N(m^*)(m^*+3)$ is about 0.235, 0.239 and 0.24 at $N = 1000$, 2000 and 4000, so it grows very slowly. These numbers are heuristic and are not used in any proof.

4 The fixed-memory limit

Corollary 4.1 (the fixed-memory limit near one)**.** *For the elephant walk with $X_1 = 1$ and fixed memory $p = (1+a)/2$, $a \in (\frac{1}{2}, 1)$, put $S_k = \sum_{j=1}^{k} X_j$ and let $\mathcal{L}_a = \lim_{k\to\infty} S_k/k^a$ as in [5, Theorem 3.7]. If Z has characteristic function $\mathbb{E}e^{itZ} = \exp\{-it \log|t| - \pi|t|/2\}$, continuously extended at $t = 0$, then*

$$\frac{1 - \mathcal{L}_{1-2\varepsilon}}{2\varepsilon} + \log(1/\varepsilon) \implies Z \qquad (\varepsilon \downarrow 0).$$

Proof. Fix $0 < \varepsilon < 1/8$, put $a = 1 - 2\varepsilon$, and use one infinite walk with $X_1 = 1$. By [5, (2.4), Appendix A], $\mathcal{M}_k = \Gamma(k) S_k / \Gamma(k+a)$ is a martingale with limit $\mathcal{L}_a$. It is Bercu's martingale

divided by $\Gamma(1+a)$. Its increments are

$$\mathcal{M}_{k+1} - \mathcal{M}_k = \frac{\Gamma(k+1)}{\Gamma(k+1+a)} \left(X_{k+1} - \frac{aS_k}{k} \right).$$

Log-convexity of gamma gives $1 \le k^a \Gamma(k)/\Gamma(k+a) \le (1+a/k)^{1-a} \le 1+1/k$. Since $S_k = k - 2M_k$, Lemma B.2 gives

$$\mathbb{E}\big[1 - a^2(S_k/k)^2\big] \le 4\varepsilon + \frac{4\mathbb{E}M_k}{k} \le 4\varepsilon(1+\log k).$$

Orthogonality of the increments therefore gives square-mean convergence and

$$\mathbb{E}|\mathcal{L}_a - \mathcal{M}_n|^2 \le C\varepsilon \sum_{k \ge n} k^{-2a}(1+\log k) \le C\varepsilon n^{1-2a}(1+\log n),$$

with C independent of $a \in [3/4, 1)$, since $2a - 1 \ge 1/2$.

Take $n = \lfloor \varepsilon^{-4/3} \rfloor$ and $x = n\varepsilon$. Then $x \to \infty$ and $x^3 \log^2 n/n = n^2\varepsilon^3 \log^2 n \to 0$, so Theorem 3.9(c) gives $Z_\varepsilon = (M_n - x\log x)/x \Rightarrow Z$. The tail bound gives $\mathbb{E}|(\mathcal{L}_a - \mathcal{M}_n)/\varepsilon|^2 = O(\varepsilon^{1/3}(1+\log(1/\varepsilon))) = o(1)$, using $n^{4\varepsilon} \to 1$. Also $|\mathcal{M}_n - n^{-a}S_n| \le n^{-a} = o(\varepsilon)$. Finally,

$$n^{-a}S_n = e^{2\varepsilon \log n}[1 - 2\varepsilon(\log x + Z_\varepsilon)],$$

$$\frac{1 - n^{-a}S_n}{2\varepsilon} + \log(1/\varepsilon) = Z_\varepsilon + o_{\mathbb{P}}(1).$$

The second equality uses $\log x = \log n + \log \varepsilon$, tightness of Z_ε, and $\varepsilon \log^2 n \to 0$. Replacing $n^{-a}S_n$ by $\mathcal{L}_a$ proves the claim. □

This is weak convergence. It gives neither a uniform local limit law nor a critical parameter for bimodality.

The total variation bound of Theorem 3.8 also controls a constrained likelihood estimator for the reversible fluctuation assay, and Theorem S4.1 of Online Resource 1 gives its concentration and mean-bias bounds.

5 Verification

The proofs use three certified computations, apart from short exact ratios such as those in Proposition 3.7. First, `certify_elephant.py` encloses the 60-term sum in Theorem 2.6(b) in $[102.586396, 102.586398]$. With the analytic tail, it is below 102.59, giving the constant 5200. Any finite bound suffices asymptotically, but this value determines the explicit thresholds in Proposition 3.7 and the second-order results. Second, the same script certifies the Landau derivative and curvature enclosures of Lemma 3.10. Lean checks these independently.

Third, `certify_convex.py` brackets each crossing for $10 \le N \le 2000$ and encloses the law throughout that bracket, checking every possible mode and the required second differences (Proposition 3.24). The separate `certify_center.py` supplies the finite endpoint-dip input of Proposition 3.7 and independently checks the center minimum. The latter checks are no longer needed for Corollary 3.19. For $5 \le N \le 9$, the stated convexity and center checks are kernel-checked in Lean. The outputs and rational witnesses are in `data/certify_convex_output_2026-10-17.txt`, `data/certify_center_output_2026-10-17.txt`, `data/witnesses_convex_2026-10-17.jsonl`, and `data/witnesses_center_2026-10-17.jsonl`. The witnesses record

script hashes and software versions. Each certification script fails if a required check fails, and range-based scripts also reject empty ranges.

The Lean 4.33.1 library in `kagey-problems/problem131/lean/PaperD` proves the finite identities, crossing bounds, sign arguments and certificate checks. The existing formal bulk and eventual-shape proofs use six named classical inputs, namely Fourier inversion, the compound Poisson characteristic function, Efron–Stein, Scheffé, unimodality of class-L laws, and Hoeffding's hypergeometric bound. Their sources and precise formal status are in `MAP.md` and `coverage_tables.tex`. The library uses Mathlib revision `0df444a360ea`; the recorded kernel build reports only `propext`, `Classical.choice` and `Quot.sound`. Generated certificate rows are checked with exact outward-rounded integer arithmetic, without `native_decide`. This does not formalize the six classical inputs or Conjecture 3.26. Lemma 3.18 and the full all-even shape in Corollary 3.19 are kernel-checked, including both complete intervals of maximizers. The composed theorem `shape_at_crossing_all_even` needs only $N = 2h$ with $h \geq 4$. The maximum-height window in Corollary 3.20, the total variation strengthening of Theorem 3.8, the likelihood theorem of Online Resource 1 (Theorem S4.1) and the fixed-memory bridge are also not yet formalized. The library's earlier pointwise compound Poisson bound remains valid.

The separate Python runs used Python 3.14.3, python-flint 0.9.0 (FLINT 3.6.0, including Arb), mpmath 1.3.0, numpy 2.4.2 and scipy 1.17.1. The figures, tables and other numerical illustrations are not proof inputs. Further checks and rerun commands are documented in the repository named under Code availability.

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A note on the use of AI

I wrote the original draft and its mathematics. I used Claude Opus 5.5 (Anthropic) for revision and proof checking, and later used OpenAI Codex to revise the exposition and check references. This included the abstract, introduction, description of the verification and the arrangement of the appendices and the Online Resource. Both models were used to expand literature search beyond what I had included, and all sources were read by me before citation if appropriate. AI tools also generated the Lean files and parts of the verification code, which I reviewed against the paper. Section 5 distinguishes the formal proofs, their named classical hypotheses and the separate numerical checks. I take responsibility for all mathematical claims, proofs and computations.

Statements and Declarations

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Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Data availability. The outputs of the computations described in Section 5 are in the folder `kagey-problems/problem131/paperD/data` of the repository named under Code availability. Online Resource 1 is the file `online_resource.pdf` in the folder `kagey-problems/problem131/paperD`, built from `online_resource.tex`.

Code availability. All code, data and Lean files supporting this paper are in the folder `kagey-problems/problem131` of https://github.com/apemm/Kagey-Problems. The supporting code, certificate records and Lean files are those at commit `6f7f9068fc0e` of that repository. To rerun the checks, run `python -B verify_all.py --paper d` in that folder, which takes several minutes. The certified computations behind Propositions 3.24 and 3.7 take about an hour and half an hour, and run separately in `paperD` as `python -B certify_convex.py 10 2000` and `python -B certify_center.py 10 2000`. The scripts need `numpy`, `scipy`, `mpmath` and `python-flint`, and the file `SUPPLEMENT.md` says what each script checks.

Use of AI tools. See the note on the use of AI tools just before these declarations.

Author contributions. I am the sole author and am responsible for the whole paper.

A Proofs for Section 2

We prove Theorem 2.6. Throughout, $N \geq 2$ is even and we condition on $X_1 = +1$. By Proposition 2.5(a) we need to bound $V_{t+1}(1)$, and we compare V with explicit hitting probabilities of Pólya urns. Let $\mathcal{L}_s F(b) = (1-r_s(b))F(b)+r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$ and $V_N(b) = \mathbf{1}\{b = h\}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from $b \geq 1$ the chain stays at states ≥ 1, and for $1 \leq b \leq s-1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h,\ 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_N = \{h\}$.

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $r_s^c(b) = \frac{b+c}{s+2c}$. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the two colors, that is, with c extra balls of each color. For $b+c > 0$ and $s-b+c > 0$ let $\Phi_s^c(b) = \binom{N-s}{h-b}\mathrm{B}(h+c, h+c)/\mathrm{B}(b+c, s-b+c)$, where B is the beta function and the binomial coefficient is 0 unless $0 \leq h-b \leq N-s$. Theorem 2.6(a) will follow from Lemma A.1(d), and (b) from Proposition A.4.

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq N$, $b+c > 0$ and $s-b+c > 0$.*

(a) The number $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time N. So $\Phi_N^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = r_s^c(b)\Phi_{s+1}^c(b+1) + (1 - r_s^c(b))\Phi_{s+1}^c(b)$ for $s < N$.

(b) If $s < N$ and $\Phi_s^c(b) > 0$, then $r_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \bar{r}_s(b) := \frac{h-b}{N-s}$.

(c) If $0 < \varepsilon < \frac{1}{2}$ and $\alpha = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = r_s^{\alpha s}(b)$ for $0 \leq b \leq s$.

(d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \alpha s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s-1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b+c$, $y = s-b+c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. As in the proof of Lemma 2.4, each order of the $N-s$ draws with $h-b$ draws of the first color has probability $(x)_{h-b}(y)_{N-s-h+b}/(x+y)_{N-s} = \mathrm{B}(h+c, h+c)/\mathrm{B}(x, y)$, and there are $\binom{N-s}{h-b}$ such orders. The recursion is the Markov property.

(b) Use $\binom{N-s-1}{h-b-1}/\binom{N-s}{h-b} = \frac{h-b}{N-s}$ and $\mathrm{B}(x,y)/\mathrm{B}(x+1,y) = \frac{x+y}{x} = 1/r_s^c(b)$.

(c) Since $1+2\alpha = 1/(1-2\varepsilon)$, we have $\frac{b+\alpha s}{s+2\alpha s} = (1-2\varepsilon)(\frac{b}{s}+\alpha) = r_s(b)$.

(d) We check the hypotheses of Lemma A.2(b) below for $F_u = \Phi_u^c$ on $\{1,\dots,u-1\}$, where $s \le u \le N$. The reason is that M has $\alpha u \ge c$ extra balls at time u, so it is pulled toward $\frac{u}{2}$ at least as strongly as the Pólya(c) chain, and from a state closer to $\frac{u}{2}$ it is easier to end at h.

By (a), $F_N(b) = \mathbf{1}\{b=h\}$, and F_u vanishes off $\mathcal{S}_u$ because the binomial coefficient does. Let $s \le u < N$ and $b \in \mathcal{S}_u$. Subtracting the recursion in (a) from the definition of $\mathcal{L}_u$ gives

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = \big(r_u(b) - r_u^c(b)\big)\big(F_{u+1}(b+1) - F_{u+1}(b)\big).$$

We show that the two factors on the right never have opposite signs. For the first factor, $r_u^c(b) = \frac{1}{2} - (\frac{u}{2}-b)/(u+2c)$, and by (c) the same formula with αu in place of c gives $r_u(b)$. So the first factor is $\frac{u}{2}-b$ times $\frac{1}{u+2c} - \frac{1}{u+2\alpha u}$. The second of these is nonnegative because $c \le \alpha s \le \alpha u$. Hence the first factor has the sign of $\frac{u}{2}-b$ or vanishes. For the second factor there are three cases. If $b=h$ then $b \ge \frac{u}{2}$ and $F_{u+1}(b+1) = 0$, so the factor is ≤ 0. If $u-b=h$ then $b \le \frac{u}{2}$ and $F_{u+1}(b) = 0$, so the factor is ≥ 0. Otherwise both values are positive, and the formula for Φ gives $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$. With $b = \frac{u}{2} - d$ the numerator minus the denominator is $2d(h+c)$, so the ratio is ≥ 1 if $b \le \frac{u}{2}$ and ≤ 1 if $b \ge \frac{u}{2}$. In all three cases the second factor has the sign of $\frac{u}{2}-b$ or vanishes. Hence $\mathcal{L}_u F_{u+1} \ge F_u$ on $\mathcal{S}_u$, and Lemma A.2(b) with $s_0 = s$ gives $V_s \ge F_s$. Finally $\Phi_{t+1}^0(1) = \beta_{t+1}(h) = \pi_t$ by (a) and Lemma 2.4. □

Lemma A.2. *Let $1 \le s_0 \le N$, and for $s_0 \le s \le N$ let $F_s \ge 0$ be a function on $\{1,\dots,s-1\}$.*

(a) If $F_N(h) \ge 1$ and $\mathcal{L}_s F_{s+1} \le K_s F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, with constants $K_s \ge 0$, then $V_s \le \big(\prod_{u=s}^{N-1} K_u\big) F_s$ on $\mathcal{S}_s$.

(b) If $F_N(b) \le \mathbf{1}\{b=h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$ for $s_0 \le s < N$, then $V_s \ge F_s$.

Proof. We use backward induction on s, from $V_N(b) = \mathbf{1}\{b=h\}$ and $V_s = \mathcal{L}_s V_{s+1}$. Note that $\mathcal{L}_s$ is an average of two values, so it keeps inequalities between functions. In (a), suppose that $V_{s+1} \le \prod_{u>s} K_u\, F_{s+1}$ on $\mathcal{S}_{s+1}$. This also holds off $\mathcal{S}_{s+1}$, because there $V_{s+1} = 0$ and $F_{s+1} \ge 0$. So on $\mathcal{S}_s$ we get $V_s = \mathcal{L}_s V_{s+1} \le \prod_{u>s} K_u\, \mathcal{L}_s F_{s+1} \le \prod_{u \ge s} K_u\, F_s$. In (b), suppose that $V_{s+1} \ge F_{s+1}$ on $\{1,\dots,s\}$. Then $V_s = \mathcal{L}_s V_{s+1} \ge \mathcal{L}_s F_{s+1} \ge F_s$ on $\mathcal{S}_s$, and $F_s = 0 \le V_s$ off $\mathcal{S}_s$. □

Idea of the proof of the upper bound. We compare M with an urn whose number of extra balls, $\kappa(s+10)$ of each color at time s, grows more than twice as fast as the αs of M. Its hitting probability $\Phi_s^{\kappa_s}$ is explicit. One step of M applied to it gives it back times a factor Δ_s, and Proposition A.4 shows that $\log \Delta_s \le \kappa/s$ up to a smaller term. Two effects enter Δ_s. Raising the extra balls from κ_s to κ_{s+1} changes Φ, and we bound this through the derivative of Φ_s^τ in τ (Lemma A.3). Also M pulls toward the middle less than the comparison urn, which helps. The product of the factors is about $e^{\kappa H_N}$. Let ψ be the digamma function.

Lemma A.3. *Let $2 \le s \le N$, $b \in \mathcal{S}_s$ and $\tau \ge 0$. Put $X = b+\tau$, $Y = s-b+\tau$, $\Sigma = X+Y$ and $z = (s-2b)^2/\Sigma^2$. Then*

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1-z) + \frac{1+z}{(1-z)\Sigma} + \frac{2(1+z)}{3\Sigma^2(1-z)^2}.$$

Proof. We differentiate the beta functions and bound the digamma function by the first terms of its expansion. Only the two beta functions in $\Phi_s^\tau(b)$ depend on τ, and $\log \mathrm{B}(x,y) = \log\Gamma(x)+\log\Gamma(y)-\log\Gamma(x+y)$. So the derivative is $2\psi(\Sigma)-\psi(X)-\psi(Y)$ minus the difference $2\psi(N+2\tau)-2\psi(h+\tau)$. The duplication formula [33, 5.5.5] gives $2\psi(2y) = \psi(y)+\psi(y+\frac12)+2\log 2$, and ψ is increasing. With $y = h+\tau$ this shows that the difference we subtract is more than $2\log 2$. For $x>0$, [33, 5.9.15] writes $\log x - \frac{1}{2x} - \psi(x)$ as $2\int_0^\infty \frac{t\,dt}{(t^2+x^2)(e^{2\pi t}-1)}$, which lies in $(0, \frac{1}{12x^2})$ since $\int_0^\infty \frac{t\,dt}{e^{2\pi t}-1} = \frac{1}{24}$. So $\psi(\Sigma) < \log\Sigma - \frac{1}{2\Sigma}$, and $\psi(x) > \log x - \frac{1}{2x} - \frac{1}{12x^2}$ for $x = X$ and $x = Y$. Hence the derivative is less than $\log\frac{\Sigma^2}{4XY} + (\frac{1}{2X}+\frac{1}{2Y}-\frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2}+\frac{1}{Y^2})$. Now $X-Y = 2b-s$, so $4XY = \Sigma^2-(X-Y)^2 = \Sigma^2(1-z)$ and $X^2+Y^2 = \Sigma^2(1+z)/2$. So the three terms are $-\log(1-z)$, $\frac{X^2+Y^2}{2XY\Sigma} = \frac{1+z}{(1-z)\Sigma}$ and $\frac{X^2+Y^2}{12X^2Y^2} = \frac{2(1+z)}{3\Sigma^2(1-z)^2}$. □

Proposition A.4. *Let $0<\varepsilon<\frac14$, $\kappa = \frac{2\varepsilon}{1-4\varepsilon}$, $\kappa_s = \kappa(s+10)$ and $\Sigma_s = s+2\kappa_s$. For $1\le s\le N-1$ and $b\in\mathcal{S}_s$ let $\Delta_s(b) = \mathcal{L}_s\Phi_{s+1}^{\kappa_{s+1}}(b)/\Phi_s^{\kappa_s}(b)$.*

(a) Let $c=\kappa_s$, $c'=\kappa_{s+1}$, $d=\frac{s}{2}-b$, $u=\frac{d}{s+2c'}$, $v=\frac{d}{N-s}$ and $\rho = \frac{s+2c'}{s(1+2\alpha)}$. Then $|u|<\frac12$ and

$$\Delta_s(b) = \mathcal{J}\,\frac{\Phi_s^{c'}(b)}{\Phi_s^{c}(b)}, \qquad \mathcal{J} = 1-(\rho-1)\frac{u(u+v)}{\frac14-u^2}.$$

(b) The factor satisfies $\log\Delta_s(b) \le \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

(c) For $1\le t\le h$, $V_{t+1}(1) \le \pi_t \exp\big(\kappa\big[H_{N-1}-H_t+\frac{2}{3t}\big] + \kappa_{t+1}\big[\log\frac{(t+1)^2}{4t}+\frac23\big]\big)$.

(d) The center satisfies $C_N \le U_N := \varepsilon c_1 e^{\kappa H_{N-1}}\sum_{t=1}^h w_t(1-\varepsilon)^{t-1}e^{\kappa\mathcal{G}(t)}$, where $\mathcal{G}(t) = (t+11)\big(\log\frac{(t+1)^2}{4t}+\frac23\big) - H_t + \frac{2}{3t}$.

The slope κ solves $2(\kappa-\alpha)/(1+2\alpha)=\kappa$, so that in (b) the gain in $\mathcal{J}$ cancels the main loss from raising the extra balls from c to c'.

Proof. (a) Write $r' = r_s^{c'}(b)$, $\bar r = \bar r_s(b)$ and $r = r_s(b)$. By Lemma A.1(b) with c', and the recursion in Lemma A.1(a), we have $r'\Phi_{s+1}^{c'}(b+1) = \bar r\,\Phi_s^{c'}(b)$ and $(1-r')\Phi_{s+1}^{c'}(b) = (1-\bar r)\Phi_s^{c'}(b)$. So the numerator of $\Delta_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar r)}{1-r'} + \frac{r\bar r}{r'}$. Here $r' = \frac12-u$, $\bar r = \frac12+v$ and $r = \frac12-\rho u$, since $\frac12 - r = (1-2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $r'(1-r') = \frac14-u^2$ the numerator is $f(u,v)+f(-u,-v)$ with $f(u,v) = (\frac12+\rho u)(\frac12-v)(\frac12-u)$, and this equals $\frac14-\rho u^2-(\rho-1)uv$. Also $|d|\le\frac{s}{2}-1$, so $|u|<\frac12$.

(b) *Step 1. The factor $\mathcal{J}$.* Write $c=\kappa_s$, $c'=c+\kappa$ and $\Sigma=\Sigma_s$. Since $c'-\alpha s = (\kappa-\alpha)s+11\kappa$, the choice of κ gives $\rho-1 = \kappa+\nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\alpha)}$. As u and v have the sign of d, $u(u+v)\ge u^2$. So $\mathcal{J}\le 1-(\kappa+\nu_s)W'$ with $z' = 4u^2$ and $W' = \frac{z'}{1-z'}$. Also $\mathcal{J}>0$ by (a), since $0<r<1$ and $\bar r$, $1-\bar r$ are not both 0. So $\log\mathcal{J}\le -(\kappa+\nu_s)W'$.

Step 2. The ratio $\Phi_s^{c'}/\Phi_s^c$. The logarithm $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau\in[c,c']$ of the derivative in Lemma A.3. Its bound increases in z and decreases in Σ, while z decreases and Σ increases in τ. So we may take the bound at $\tau=c$, where $X+Y=\Sigma$, and multiply it by the length κ of the interval. Put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1+W$, $\frac{1+z}{1-z} = 1+2W$ and $-\log(1-z)\le W$, so

$$\log\Delta_s(b) \le \kappa\Big[W + \frac{1+2W}{\Sigma} + \frac{2(1+2W)(1+W)}{3\Sigma^2}\Big] - (\kappa+\nu_s)W'. \tag{3}$$

Step 3. What is left to show. The terms of (3) without W and W' are $\frac{\kappa}{\Sigma}+\frac{2\kappa}{3\Sigma^2}$, which is the bound we want. Since $(1+2W)(1+W)=1+3W+2W^2$, the other terms are

$$\kappa(W-W')+\frac{2\kappa W}{\Sigma}+\frac{2\kappa W}{\Sigma^2}+\frac{4\kappa W^2}{3\Sigma^2}-\nu_s W',$$

and we must show that they add up to at most 0. If $W=0$ then $d=0$, so $W'=0$ and there is nothing to do. So let $W>0$. The first four terms are positive, and we show that each is a small multiple of $\frac{\kappa}{s}W$. The last term is negative (it comes from the shift 10 in κ_s), and it is a larger multiple of $\frac{\kappa}{s}W$. We need three facts, with $X=b+c$ and $Y=s-b+c$ as in Step 2.

(i) We have $1+W\le\frac{\Sigma}{2(1+c)}$. Indeed $X,Y\ge 1+c$, since $b\ge 1$ and $s-b\ge 1$, and the larger of X and Y is at least $\frac{\Sigma}{2}$. So $XY\ge(1+c)\Sigma/2$, and $1+W=\frac{1}{1-z}=\frac{\Sigma^2}{4XY}$.

(ii) We have $W-W'\le\frac{2\kappa}{1+c}W$, and so $W'\ge\frac{9}{11}W$. Indeed z and z' have the same numerator $(s-2b)^2$, and their denominators are Σ^2 and $(s+2c')^2=(\Sigma+2\kappa)^2$. So $z'=\lambda z$ with $\lambda=(\frac{\Sigma}{\Sigma+2\kappa})^2\ge 1-\frac{4\kappa}{\Sigma}$. Then

$$W-W'=\frac{(1-\lambda)z}{(1-z)(1-\lambda z)}\le(1-\lambda)W(1+W)\le\frac{4\kappa}{\Sigma}\,W\,\frac{\Sigma}{2(1+c)}=\frac{2\kappa}{1+c}W,$$

where we used $1-\lambda z\ge 1-z$ and then (i). Since $c=\kappa(s+10)\ge 11\kappa$, we have $\frac{2\kappa}{1+c}\le\frac{2}{11}$.

(iii) We have $\Sigma\ge s$, $1+c>\kappa s$ and $\Sigma^2\ge 3s$. The first two are clear. For the third, $W>0$ means $s\ne 2b$, and with $b\ge 1$ and $s-b\ge 1$ this forces $s\ge 3$.

Step 4. The terms with W. We bound the four positive terms one by one.

$$\begin{aligned}
\kappa(W-W')&\le\frac{2\kappa^2}{1+c}W\le\frac{2\kappa}{s}W && \text{by (ii) and } 1+c>\kappa s,\\
\frac{2\kappa W}{\Sigma}&\le\frac{2\kappa}{s}W && \text{by } \Sigma\ge s,\\
\frac{2\kappa W}{\Sigma^2}&\le\frac{2\kappa}{3s}W && \text{by } \Sigma^2\ge 3s,\\
\frac{4\kappa W^2}{3\Sigma^2}&\le\frac{4\kappa W(1+W)}{3\Sigma^2}\le\frac{2\kappa W}{3\Sigma(1+c)}\le\frac{2\kappa}{3s}W && \text{by (i) and } \Sigma\ge s.
\end{aligned}$$

The four bounds add up to $\frac{16\kappa}{3s}W$. For the negative term, (ii) gives $-\nu_s W'\le-\frac{9}{11}\nu_s W=-\frac{18\kappa}{s(1+2\alpha)}W$. So

$$\log\Delta_s(b)\le\frac{\kappa}{\Sigma}+\frac{2\kappa}{3\Sigma^2}+W\Big[\frac{16\kappa}{3s}-\frac{18\kappa}{s(1+2\alpha)}\Big],$$

and the bracket is negative because $1+2\alpha<2$ for $\varepsilon<\frac14$, so $18/(1+2\alpha)>9>16/3$.

(c) We first compare $V_{t+1}(1)$ with $\Phi_{t+1}^{\kappa_{t+1}}(1)$, and then this with $\pi_t=\Phi_{t+1}^0(1)$. For the first step apply Lemma A.2(a) with $s_0=t+1$, $F_s=\Phi_s^{\kappa_s}$ and $K_s=\exp(\frac{\kappa}{s}+\frac{2\kappa}{3s^2})$. This is allowed, since $F_N(h)=1$, $F_s>0$ on $\mathcal{S}_s$, and $\mathcal{L}_sF_{s+1}=\Delta_sF_s\le K_sF_s$ on $\mathcal{S}_s$ by (b) and $\Sigma_s\ge s$. The product of the K_s over $t<s<N$ is the exponential of $\kappa(H_{N-1}-H_t)+\frac{2\kappa}{3}\sum_{s>t}s^{-2}$, and $\sum_{s>t}s^{-2}\le\frac1t$. Since $1\in\mathcal{S}_{t+1}$, we get $V_{t+1}(1)\le e^{\kappa[H_{N-1}-H_t+2/(3t)]}\,\Phi_{t+1}^{\kappa_{t+1}}(1)$.

For the second step, $\log(\Phi_{t+1}^{\kappa_{t+1}}(1)/\pi_t)$ is the integral over $\tau\in[0,\kappa_{t+1}]$ of the derivative in Lemma A.3 with $s=t+1$ and $b=1$. As in Step 2 of (b), the bound of that lemma is largest at $\tau=0$. There $X=1$, $Y=t$ and $\frac{1}{1-z}=\frac{(t+1)^2}{4t}$, and the bound is $\log\frac{(t+1)^2}{4t}+\frac{t^2+1}{2t(t+1)}+\frac{t^2+1}{12t^2}\le\log\frac{(t+1)^2}{4t}+\frac23$, because the two fractions are at most $\frac12$ and $\frac16$ for $t\ge1$. Multiplying by the length κ_{t+1} of the interval gives (c).

(d) Insert (c) into Proposition 2.5(a), and use $\kappa_{t+1}=\kappa(t+11)$ and $\pi_t=c_1w_t$. □

Proof of Theorem 2.6. (a) By Proposition 2.5(a) and Lemma A.1(d), $C_N \ge \varepsilon c_1 \sum_t w_t (1-\varepsilon)^{t-1}$. Now $(1-\varepsilon)^{t-1} \ge 1-(t-1)\varepsilon$ and $\sum_t w_t(t-1) = 2 - \frac{6}{h+2}$ by Proposition 2.5(c). So $C_N \ge \varepsilon c_1 (1-2\varepsilon)$, and $c_1 = \frac{4}{N+2}$.

(b) *Step 1. The constants.* Let $0 < \varepsilon(1+\log N) \le \frac{1}{100}$. Since $\log N \ge 0$, we have $\varepsilon \le \frac{1}{100}$. So $1-4\varepsilon \ge 0.96$ and $\kappa \le 2\varepsilon/0.96 \le 2.1\varepsilon$. Also $\kappa \le 2\varepsilon(1+8\varepsilon)$, because $(1+8\varepsilon)(1-4\varepsilon) = 1+4\varepsilon(1-8\varepsilon) \ge 1$. Since $\log \frac{(t+1)^2}{4t} \ge 0$ and $\frac{2}{3}(t+11) > 1 + \log t \ge H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \le \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \ge \frac{2}{3t}$, also $\mathcal{G}(t) \le (t+11)\log(t+1) \le 12t\log(t+1)$. So $\kappa\mathcal{G}(t) \le 25.2\,\varepsilon t \log(t+1)$, and this is at most $\frac{t}{2}\log 2$ for $t \le h$, since then $25.2\,\varepsilon \log(t+1) \le 25.2\,\varepsilon \log N \le 0.252 < \frac{1}{2}\log 2$.

Step 2. The sum over t. Since $e^y - 1 \le y e^y$ for $y \ge 0$, Step 1 gives $e^{\kappa\mathcal{G}(t)} - 1 \le \kappa \mathcal{G}(t) 2^{t/2}$. By Proposition A.4(d), $(1-\varepsilon)^{t-1} \le 1$, $\sum_t w_t = 1$ and $w_t \le 2t2^{-t}$ (Proposition 2.5(c)),

$$C_N \le \varepsilon c_1 e^{\kappa H_{N-1}} \Big(1 + 50.4\,\varepsilon \sum_{t \ge 1} t^2 \log(t+1) 2^{-t/2}\Big),$$

and we bound the series as follows. The ratio of its consecutive terms is $(1+\frac{1}{t})^2 \frac{\log(t+2)}{\log(t+1)} 2^{-1/2}$, which decreases in t and is below $\frac{3}{4}$ at $t = 60$. So the terms with $t > 60$ add up to less than four times the term at $t = 61$, and this is below 10^{-4}. The first 60 terms add up to 102.5864 to four decimals (enclosed in ball arithmetic by `certify_elephant.py`), so the series is below 102.59.

Step 3. The factor $e^{\kappa H_{N-1}}$. Finally $H_{N-1} \le 1 + \log N$, so $\kappa H_{N-1} \le 2\varepsilon \log N + 2\varepsilon + 16\varepsilon^2(1 + \log N) \le 2\varepsilon \log N + 3\varepsilon$. With $1 + y \le e^y$ this proves (b), since $3 + 50.4 \cdot 102.59 < 5200$. Any upper bound for the series below $(5200-3)/50.4$, which exceeds 103.1, would give the same constant. □

B The families and the mode

This appendix states the family estimates used for the mode bound in Corollary 3.11. Their proofs are in Appendix D, in the order stated here, and also support the tail theorem of Online Resource 1. We keep the conventions of Section 3. All constants in this appendix are explicit. Appendix D writes C for an absolute constant whose value may change from line to line. Since M is independent of X_1, $f_N(m) = \mathbb{P}(M_N = m)$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, and we use the paths of Y and $Z^{(j)}$ fixed after Lemma 3.3. The size $|\mathcal{T}_v|$ of the family has the law β_v of Lemma 2.4. Let $\bar\beta_v(s) = \mathbb{P}(|\mathcal{T}_v| \ge s)$, with $\bar\beta_v(s) = 1$ for $s \le 1$. By Pascal's rule $\binom{N-s}{v-1} - \binom{N-s-1}{v-1} = \binom{N-s-1}{v-2}$ the tail of β_v telescopes. So, for $1 \le s \le N-1$,

$$\bar\beta_v(s) = \binom{N-s}{v-1} \Big/ \binom{N-1}{v-1} = \prod_{i=1}^{v-1} \Big(1 - \frac{s-1}{N-i}\Big) \le e^{-(v-1)(s-1)/(N-1)}, \qquad \beta_v(s) = \frac{v-1}{N-s}\,\bar\beta_v(s), \tag{4}$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{N-d}, \qquad \frac{\bar\beta_v(s-1)}{\bar\beta_v(s)} = 1 + \frac{v-1}{N-s-v+2}. \tag{5}$$

Lemma B.1 (monotone sandwich)**.** *Let $0 \le a \le b \le N$. If $m_h \le a$, then $(b-a+1) f_N(b) \le \mathbb{P}(a \le M_N \le b)$. If $b \le m_h$, then $(b-a+1) f_N(a) \le \mathbb{P}(a \le M_N \le b)$.*

Lemma B.2 (tail)**.** *Let $0 < \varepsilon < 1$, $k_0 \ge 1$ and $t \ge k_0$. For every integer $z \ge 1$,*

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \frac{\varepsilon t H_z}{z}, \qquad \mathbb{E} M_t^{[k_0]} \le \varepsilon t \log \frac{t}{k_0}.$$

In particular $\mathbb{P}(M_N \ge z) \le x H_z/z$, and $\mathbb{P}(Z_s^{(j)} \ge z) \le x H_z/z$ for $j - 1 \le s \le N$.

Lemma B.3 (the first flipped family)**.** *For* $1 \le d \le N-1$,

$$\frac{x}{d(d+1)}\Big(1-\frac{2x(N-d-1)}{N(d+2)}\Big) \le \mathbb{P}\big(|\mathcal{T}_V| = d\big) \le \frac{x}{d(d+1)}, \qquad \mathbb{P}\big(|\mathcal{T}_V| \ge d\big) \le x\Big(\frac{1}{d}-\frac{1}{N}\Big).$$

Proposition B.4 (local lower bound)**.** *Let* $N \ge 64$, $0 < \varepsilon < 1$ *and* $2 \le m \le N/4$. *Then* $f_N(m) \ge \frac{x}{m(m+1)}(1-e_1'(m)) \ge \frac{x}{m(m+1)}(1-e_1(m))$, *where*

$$e_1'(m) = 2\varepsilon H_{m+1} + \frac{xH_m}{m} + \frac{2x}{m+2} + 8\varepsilon \log N + 20me^{-m/9}$$

and

$$e_1(m) = 20\Big(\frac{x\log(m+1)}{m} + \varepsilon \log N + me^{-m/9}\Big).$$

Idea of the proof. We use the formula of Lemma 3.3. Given the time j of the first flip, the count M_N is $d - Y_d + Z_{N-d}^{(j)}$, where d is the size of the first flipped family. As d grows by one this quantity moves by at most one, so it hits m at some d close to m unless Y or Z is large. (If no step of the family flips back and the rest of the tree has no flip, then $Y = Z = 0$ and the quantity is d itself, which hits m at $d = m$.) The law β_j of d is nearly flat near m, and this gives almost the full weight $\beta_j(m)$. Then we sum over j. The error e_1 collects the chance that Y or Z is large, the small slope of β_j, and the first flips that come too late to be treated this way.

Lemma B.5 (mode bound)**.** *Let* $N \ge 64$ *and* $x \ge 1$, *and put* $\bar{c} = \frac{1}{40}$ *and* $A_0 = 15400$. *If* $\varepsilon \log N \le \bar{c}$ *and* $804x\log(x+2) \le N$, *then* $m_h \le A_0 x \log^2(x+1)$. *In particular this holds if* $\varepsilon \log N \le \bar{c}$ *and* $A_0 x \log(x+2) \le N/4$.

C Proofs for the bulk

We prove Theorems 3.8 and 3.9, Lemma 3.10 and Corollary 3.11. We keep the notation of Section 3 and of Appendix B. So $\mathcal{T}$ is the tree of Lemma 3.1, $\mathcal{T}_v$ is the family of v, and μ_v is the mark of v. By Lemma 2.4 and the discussion after Lemma 3.1, $|\mathcal{T}_v|$ has the law β_v, with mean N/v, and $\sum_{v=2}^N \beta_v(d) = N/(d(d+1))$ for $1 \le d \le N-1$. Let

$$\Psi_N(\theta) = \sum_{d=1}^{N-1} \frac{e^{i\theta d}-1}{d(d+1)}, \qquad \Psi_\infty(\theta) = \sum_{d \ge 1} \frac{e^{i\theta d}-1}{d(d+1)}.$$

The law CP_N has the characteristic function $\widehat{\mathrm{CP}}_N(\theta) = e^{x\Psi_N(\theta)}$.

C.1 Proof of Theorem 3.8

We use $\mathrm{TV}(P,Q) = \frac{1}{2}\sum_k |P(k)-Q(k)|$. The case $N = 1$ is immediate. Assume $N \ge 2$ and $0 < \varepsilon < 1$. Put $Z_+ = \sum_{v=2}^N \mu_v |\mathcal{T}_v|$. Unless two marked vertices are strictly nested, $M_N = Z_+$. The family means therefore give

$$\mathbb{P}(M_N \ne Z_+) \le \varepsilon^2 \sum_{v=2}^N (N/v - 1) = \varepsilon^2(NH_N - 2N + 1).$$

Given the tree, replace the independent Bernoulli marks by independent Poisson variables of mean ε. The scalar total variation is $\varepsilon(1-e^{-\varepsilon}) \le \varepsilon^2$, since the only positive Bernoulli-minus-Poisson

mass is at 1. Product coupling and the weighted sum cost at most $(N-1)\varepsilon^2$. The resulting conditional law is $\mathrm{CP}(\Lambda)$, where $\Lambda = \varepsilon \sum_{v=2}^{N} \delta_{|\mathcal{T}_v|}$ has deterministic mass $\lambda = (N-1)\varepsilon$. By the family identity, $\mathbb{E}\Lambda\{d\} = x/[d(d+1)]$, so $\mathrm{CP}(\mathbb{E}\Lambda) = \mathrm{CP}_N$.

Use the L1 norm on signed measures and convolution $*$, for which $\|\sigma * \tau\|_1 \le \|\sigma\|_1 \|\tau\|_1$. On intensities of mass λ, the CP map and its derivatives in a zero-mass direction D are

$$F(\nu) = e^{-\lambda} \sum_{r \ge 0} \frac{\nu^{*r}}{r!}, \qquad F'(\nu)[D] = F(\nu) * D, \qquad F''(\nu)[D, D] = F(\nu) * D * D,$$

where $\nu^{*0} = \delta_0$. The series and derivatives converge in L1. Let $\mathcal{F}_j = \sigma(U_1, \ldots, U_j)$, $\Lambda_j = \mathbb{E}(\Lambda \mid \mathcal{F}_j)$ and $D_j = \Lambda_j - \Lambda_{j-1}$. Each Λ_j is nonnegative of mass λ. Along the segment between consecutive intensities F has norm 1, so Taylor's remainder has norm at most $\|D_j\|_1^2/2$. The linear term has conditional mean zero given $\mathcal{F}_{j-1}$. Telescoping expectations and dividing by two gives

$$\mathrm{TV}(\mathbb{E}F(\Lambda), F(\mathbb{E}\Lambda)) \le \frac{1}{4} \sum_{j=1}^{N-1} \mathbb{E}\|D_j\|_1^2.$$

If $\Lambda^{(j)}$ resamples only U_j, independence gives $D_j = \mathbb{E}(\Lambda - \Lambda^{(j)} \mid \mathcal{F}_j)$. Conditional Jensen thus bounds its squared norm in expectation by that of $\Lambda - \Lambda^{(j)}$. Moving the subtree of $j+1$ changes family sizes only along the old and new parent paths, excluding the root. Each altered atom contributes at most 2ε in L1. Writing dp for depth, and using $(a+b)^2 \le 2(a^2+b^2)$, gives

$$\mathbb{E}\|\Lambda - \Lambda^{(j)}\|_1^2 \le 16\varepsilon^2 \mathbb{E}\,\mathrm{dp}(U_j)^2 \le 16\varepsilon^2 (H_{j-1}^2 + H_{j-1}).$$

For the last inequality, the depth pgf satisfies $g_1(s) = 1$ and $j g_{j+1}(s) = s \sum_{v \le j} g_v(s)$. Subtraction gives $g_v(s) = \prod_{i=1}^{v-1} (i - 1 + s)/i$. Thus the depth has the law of a sum of independent Bernoulli$(1/i)$ variables, and its second moment is at most $H_{v-1}^2 + H_{v-1}$. Average over the independent uniform parent $U_j \in \{1, \ldots, j\}$.

Adding the three errors proves the stronger bound

$$\mathrm{TV}(\mathcal{L}(M_N), \mathrm{CP}_N) \le \varepsilon^2 \left\{ N H_N - N + 4 \sum_{j=0}^{N-2} (H_j^2 + H_j) \right\} \le \delta_\Sigma.$$

Finally $H_N \le 1 + \log N$ and $H_j \le H_N$ give $\delta_\Sigma \le (x^2/N)\{\log N + 2 + 4(H_N^2 + H_N)\} \le \delta_B$. Every singleton discrepancy is at most total variation. □

C.2 Proof of Theorem 3.9

We write $B(t) = -it \log|t| - \frac{\pi}{2}|t|$ for real $t \ne 0$, and $B(0) = 0$. For $t > 0$ this is the value of $u \log u$ at $u = -it$, with the principal logarithm, and $B(-t) = \overline{B(t)}$. We need three facts about Ψ_N and Ψ_∞.

Fact 1. For real θ, $|\Psi_N(\theta) - \Psi_\infty(\theta)| \le \sum_{d \ge N} \frac{2}{d(d+1)} = \frac{2}{N}$.

Fact 2. For $0 < |\theta| \le \pi$ we have $\Psi_\infty(\theta) = -i\theta \log(-i\theta) + R(\theta)$ with $|R(\theta)| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big)$. The reason is that Ψ_∞ has a closed form in which θ enters only through $1 - e^{i\theta}$, and $1 - e^{i\theta}$ is close to $-i\theta$ for small θ. The term R is what this replacement changes. We prove it in the next four paragraphs.

We start with the closed form. For $0 < |s| \le 1$, $s \ne 1$,

$$\sum_{d\ge1} \frac{s^d-1}{d(d+1)} = \chi(s) := \frac{(1-s)\log(1-s)}{s},$$

because $\frac{1}{d(d+1)} = \frac{1}{d} - \frac{1}{d+1}$, $\sum_d s^d/d = -\log(1-s)$, $\sum_d s^d/(d+1) = (-\log(1-s)-s)/s$ and $\sum_d \frac{1}{d(d+1)} = 1$. So $\Psi_\infty(\theta) = \chi(e^{i\theta})$.

Next we compare $1-e^{i\theta}$ with $-i\theta$. Write $1-e^{i\theta} = (-i\theta)\,\mathsf{u}$ with $\mathsf{u} = \operatorname{sinc}(\theta/2)\,e^{i\theta/2}$, where $\operatorname{sinc} y = \sin y/y$ is positive for $|y| \le \frac{\pi}{2}$. So u is close to 1 when θ is small. The argument of $-i\theta$ is $\mp\frac{\pi}{2}$ for $\pm\theta > 0$ and that of u is $\theta/2$. They add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$, so $\log(1-e^{i\theta}) = \log(-i\theta) + \log \mathsf{u}$ for principal logarithms.

Now we get a formula for R. By the two steps above, $\chi(e^{i\theta}) = (-i\theta)\,\mathsf{u}e^{-i\theta}\big(\log(-i\theta) + \log \mathsf{u}\big)$. Subtracting $-i\theta\log(-i\theta)$ gives

$$R(\theta) = -i\theta\log(-i\theta)\big(\mathsf{u}e^{-i\theta}-1\big) - i\theta\,\mathsf{u}e^{-i\theta}\log \mathsf{u}.$$

It remains to bound the three quantities in this formula. We show that

$$|\mathsf{u}e^{-i\theta}-1| \le 0.631\,|\theta|, \qquad |\log \mathsf{u}| \le 0.53\,|\theta|, \qquad |\log(-i\theta)| \le |\log|\theta|| + \frac{\pi}{2}.$$

For the first, put $y = \theta/2$. Then $|\mathsf{u}e^{-i\theta}-1| = |\operatorname{sinc} y - e^{iy}| \le (1-\operatorname{sinc} y) + |1-e^{iy}| \le \frac{y^2}{6} + |y| \le (\frac{1}{2} + \frac{\pi}{24})|\theta|$, where the last step uses $|\theta| \le \pi$, and $\frac{1}{2} + \frac{\pi}{24} \le 0.631$. For the second, $|\log \mathsf{u}|^2 = \log^2 \operatorname{sinc}(\theta/2) + \theta^2/4$, and $-\log\operatorname{sinc} y = \sum_{k\ge1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log\operatorname{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and its value at $|\theta| = \pi$ gives $|\log \mathsf{u}| \le \big(\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2}\big)^{1/2}|\theta| \le 0.53\,|\theta|$. The third holds because $-i\theta$ has modulus $|\theta|$ and argument $\pm\frac{\pi}{2}$. Since $|\mathsf{u}e^{-i\theta}| \le 1$, the formula for R now gives

$$|R(\theta)| \le |\theta|\Big(|\log|\theta|| + \frac{\pi}{2}\Big)\cdot 0.631\,|\theta| + |\theta|\cdot 0.53\,|\theta| \le \theta^2\big(0.631\,|\log|\theta|| + 1.53\big),$$

as $0.631\cdot\frac{\pi}{2} + 0.53 \le 1.53$.

Fact 3. For $|\theta| \le \pi$, $-\operatorname{Re}\Psi_\infty(\theta) = \sum_d \frac{1-\cos\theta d}{d(d+1)} \ge \frac{|\theta|}{2\pi}$. We may take $\theta \ne 0$. Keep only the terms with $d \le D := \lfloor \pi/|\theta| \rfloor$ and use $1-\cos v \ge 2v^2/\pi^2$ for $|v| \le \pi$. This gives at least $\frac{2\theta^2}{\pi^2}\sum_{d\le D}\frac{d}{d+1} \ge \frac{\theta^2 D}{\pi^2} \ge \frac{|\theta|}{2\pi}$, since $D \ge \frac{\pi}{2|\theta|}$.

The function e^B is a characteristic function of class L. Let S_x have the law CP_∞ with parameter x, namely the law of $\sum_{d\ge1} d\,\mathcal{N}_d$. Its characteristic function is $e^{x\Psi_\infty(\theta)}$. By Fact 2, for $t \ne 0$ and $x \ge \max(1, |t|/\pi)$,

$$x\Psi_\infty(t/x) - it\log x = B(t) + xR(t/x), \qquad |xR(t/x)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big). \quad (6)$$

So the characteristic function of $(S_x - x\log x)/x$ tends to $e^{B(t)}$ as $x \to \infty$, for every real t. The limit is continuous at 0, so by Lévy's continuity theorem it is the characteristic function of a probability law. It is integrable, so this law has the bounded continuous density f_L of Theorem 3.9. For $0 < c < 1$ and real t we have $B(ct) = cB(t) - ict\log c$, and hence

$$e^{B(t)} = e^{B(ct)}\cdot e^{B((1-c)t)}\,e^{it[(1-c)\log(1-c) + c\log c]}.$$

The last two factors form the characteristic function of a scaled and shifted copy of the law. So the law is of class L (self-decomposable), and f_L is unimodal by [27].

Proof of (a). By Fourier inversion on $\mathbb{Z}$ and the substitution $\theta = t/x$,

$$x\,\mathrm{CP}_N(m) = \frac{1}{2\pi}\int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)}\,dt, \qquad A(t) = x\Psi_N(t/x) - it\log x.$$

Let $|t| \le \pi x$. By (6) and Fact 1,

$$|A(t) - B(t)| \le \frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N},$$

and by Facts 1 and 3, $\max(\mathrm{Re}\,A(t), \mathrm{Re}\,B(t)) \le -\frac{|t|}{2\pi} + \frac{2x}{N}$. Now $|e^A - e^B| \le |A-B|\,e^{\max(\mathrm{Re}\,A, \mathrm{Re}\,B)}$, and $\frac{2x}{N} \le \frac{1}{2}$ because $N \ge 4x$. So for $|t| \le \pi x$

$$|e^{A(t)} - e^{B(t)}| \le e^{1/2} e^{-|t|/2\pi}\Big[\frac{t^2}{x}\big(0.631(|\log|t|| + \log x) + 1.53\big) + \frac{2x}{N}\Big].$$

For $s \ge 1$ the tangent of the logarithm at 16 gives $\log s \le s/16 + \log 16 - 1$, and for $0 < s < 1$ we have $s|\log s| \le 1/e$. So $s^2|\log s| \le s^3/16 + (4\log 2 - 1)s^2 + s/e$ for every $s \ge 0$. We insert this with $s = |t|$ and integrate, using $\int_0^\infty t^k e^{-t/2\pi}\,dt = k!\,(2\pi)^{k+1}$. This gives

$$\frac{1}{2\pi}\int_{-\pi x}^{\pi x} |e^{A(t)} - e^{B(t)}|\,dt \le e^{1/2}\Big[\frac{K_0 + 16\pi^2 \cdot 0.631\log x}{x} + \frac{4x}{N}\Big],$$

where $K_0 = 6\pi^3 \cdot 0.631 + 16\pi^2\big(0.631(4\log 2 - 1) + 1.53\big) + 4\pi \cdot 0.631/e \le 538.6$. Beyond πx we use $|e^{B(t)}| = e^{-\pi|t|/2}$, which gives $\frac{1}{2\pi}\int_{|t|>\pi x} |e^{B(t)}|\,dt = \frac{2}{\pi^2} e^{-\pi^2 x/2} \le \frac{4}{\pi^4 x} \le \frac{0.042}{x}$, since $e^{-y} \le 1/y$ for $y > 0$. Since $f_L(\lambda_m) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-i\lambda_m t} e^{B(t)}\,dt$, the error is at most the sum of the two integrals. With $e^{1/2} \le 1.6488$ and $16\pi^2 \cdot 0.631 \le 99.66$ this sum is at most

$$\frac{1.6488 \cdot 538.6 + 0.042 + 1.6488 \cdot 99.66\log x}{x} + \frac{6.5952\,x}{N} \le \frac{889 + 165\log x}{x} + \frac{6.6\,x}{N}.$$

The second bound in (a) holds because $165 \le 889$. This proves (a). Fourier inversion on $\mathbb{Z}$ holds at every integer m, and $\mathrm{CP}_N(m) = 0$ for $m < 0$, so the proof covers every integer m.

Proof of (b). Since $\varepsilon \le \frac{1}{4}$, we have $N \ge 4x$. By Theorem 3.8, $|x f_N(m) - x\,\mathrm{CP}_N(m)| \le x\delta_\Sigma \le x\delta_B$. Add (a).

Proof of (c). We drop the index n. Let $x \to \infty$ and $x^3 \log^2 N/N \to 0$. Then $N \ge x \to \infty$, so $\log N \ge 1$ for large n, and $x/N \to 0$. Hence for large n we have $\varepsilon \le \frac{1}{4}$ and $x \ge 1$, and (b) applies. Its bound tends to 0, since $x\delta_B = O(x^3 \log^2 N/N)$. Only large n matter for convergence in law. Let U be uniform on $[0,1)$ and independent of M_N. Then $(M_N + U - x\log x)/x$ has the density q_N with $q_N(\lambda) = x f_N(m)$ for $\lambda_m \le \lambda < \lambda_m + \frac{1}{x}$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $q_N \to f_L$ uniformly on compact sets. By Scheffé's lemma $q_N \to f_L$ in L^1, and since $U/x \to 0$ this gives the convergence in law. At the crossing $x_N \sim 2\log N$ by Theorem 2.7, so both conditions hold. □

C.3 Proof of Lemma 3.10

Step 1. A real integral. We claim that for every real λ and $k \ge 0$,

$$f_L^{(k)}(\lambda) = \frac{1}{\pi}\int_0^\infty (-s)^k\, e^{-s\log s - \lambda s}\sin(\pi s)\,ds. \tag{7}$$

For $k=0$ this is [20, Eq. (4.3)], and we include a short proof. Since $B(-t)=\overline{B(t)}$, the definition of f_L gives $f_L(\lambda)=\frac{1}{\pi}\operatorname{Re}\int_0^\infty \Phi(-it)\,dt$ with $\Phi(u)=\exp(u\log u+\lambda u)$. The function Φ is holomorphic off the negative real axis. Put $u=\rho e^{i\varphi}$ with $-\pi<\varphi\le -\frac{\pi}{2}$. Then $\operatorname{Re}(u\log u)=\rho(\log\rho\cos\varphi-\varphi\sin\varphi)$, where $\cos\varphi\le 0$ and $\varphi\sin\varphi\ge 0$. For $\log\rho\ge|\lambda|+1$ this gives $\operatorname{Re}(u\log u+\lambda u)\le -\rho(|\cos\varphi|+\varphi\sin\varphi)\le -\rho$, because $|\cos\varphi|+\varphi\sin\varphi\ge 1$ on $[-\pi,-\frac{\pi}{2}]$ (for $\psi=\varphi+\pi\in[0,\frac{\pi}{2}]$ it equals $\cos\psi+(\pi-\psi)\sin\psi\ge\cos\psi+\sin\psi\ge 1$). Also Φ is bounded near 0. So by Cauchy's theorem we may turn the ray $u=-it$ to the lower side of the negative real axis, where $u=-s$ and $\log u=\log s-i\pi$. There $u\log u=-s\log s+i\pi s$, and $dt=i\,du=-i\,ds$. Hence

$$f_L(\lambda)=\frac{1}{\pi}\operatorname{Re}\Big[-i\int_0^\infty e^{-s\log s-\lambda s}\,e^{i\pi s}\,ds\Big]=\frac{1}{\pi}\int_0^\infty e^{-s\log s-\lambda s}\sin(\pi s)\,ds.$$

The integrand decays faster than every exponential, so we may differentiate under the integral sign. This proves (7). Also f_L'' is bounded on the real line, because $t^2e^{B(t)}$ is integrable and we may differentiate the Fourier formula for f_L twice.

Step 2. Certified enclosures. Let $-0.223\le\lambda\le 0$ and $k\in\{1,2\}$, and split the integral in (7) at $\delta=2^{-20}$ and at 30. On $[0,\delta]$ we have $-s\log s\le 1/e$, $-\lambda s\le 0.223\,s$ and $|\sin\pi s|\le\pi s$, so this part is at most $\pi e^{1/e+0.223\delta}\delta^{k+2}/(k+2)$ in absolute value. On $[30,\infty)$ we have $s\log s+\lambda s\ge cs$ with $c=\log 30-0.223>3$, so this part is at most $\int_{30}^\infty s^2e^{-cs}\,ds\le 30^2e^{-30c}$. Both are below 10^{-17}. On $[\delta,30]$ the integrand is holomorphic in a neighborhood of the path, and we enclose the integral with the rigorous integration algorithm of [28], in ball arithmetic with 128 bits. The parameter λ enters as a ball, so one run encloses the integral for every λ in an interval. The script `certify_elephant.py` does this and reports

$$f_L'(-0.2229)\in[9.232\cdot10^{-6},9.234\cdot10^{-6}],\qquad f_L'(-0.2227)\in[-6.548\cdot10^{-6},-6.546\cdot10^{-6}],$$

and $f_L''(\lambda)\in[-0.0792,-0.0786]$ for every $\lambda\in[-0.2229,-0.2227]$. In the same way it finds that f_L' is positive at -0.22278299 and negative at -0.22278297. The two enclosures are $6.8984026\cdot10^{-10}$ and $-8.8809315\cdot10^{-10}$ with radii below 10^{-17}, and these are the smallest margins in the proof.

Step 3. The mode. Write $[a,b]=[-0.2229,-0.2227]$. By Step 2, f_L' is strictly decreasing on $[a,b]$ with $f_L'(a)>0>f_L'(b)$. So f_L' has exactly one zero λ_{L} in $[a,b]$, and f_L is strictly increasing on $[a,\lambda_{\mathrm{L}}]$ and strictly decreasing on $[\lambda_{\mathrm{L}},b]$. Since f_L is unimodal, there is a point λ_* such that f_L is nondecreasing on $(-\infty,\lambda_*]$ and nonincreasing on $[\lambda_*,\infty)$. If $\lambda_*>\lambda_{\mathrm{L}}$, then f_L would be nondecreasing just to the right of λ_{L}, which is false. In the same way $\lambda_*<\lambda_{\mathrm{L}}$ is impossible. So $\lambda_*=\lambda_{\mathrm{L}}$, and f_L is nondecreasing on $(-\infty,\lambda_{\mathrm{L}}]$ and nonincreasing on $[\lambda_{\mathrm{L}},\infty)$. By Step 2 again, $\lambda_{\mathrm{L}}\in(-0.22278299,-0.22278297)$. This proves (a) and (b).

Step 4. The constant κ_L. Let $\delta_0=\min(\lambda_{\mathrm{L}}-a,b-\lambda_{\mathrm{L}})>0$. For $|\lambda-\lambda_{\mathrm{L}}|\le\delta_0$, Taylor's formula with $f_L'(\lambda_{\mathrm{L}})=0$ and $f_L''\le -0.078$ on $[a,b]$ gives $f_L(\lambda_{\mathrm{L}})-f_L(\lambda)\ge 0.039(\lambda-\lambda_{\mathrm{L}})^2$. For $|\lambda-\lambda_{\mathrm{L}}|>\delta_0$ the monotonicity of Step 3 gives $f_L(\lambda)\le f_L(\lambda_{\mathrm{L}}\pm\delta_0)\le f_L(\lambda_{\mathrm{L}})-0.039\,\delta_0^2$. Since $\delta_0<1$, the bound in (c) holds with $0.039\,\delta_0^2$ in place of κ_L. By (a), $\lambda_{\mathrm{L}}-a>0.2229-0.22278299>1.17\cdot10^{-4}$ and $b-\lambda_{\mathrm{L}}>0.22278297-0.2227=8.297\cdot10^{-5}$, so $0.039\,\delta_0^2>0.039\cdot(8.297\cdot10^{-5})^2>2.68\cdot10^{-10}$. So (c) holds with $\kappa_L=2.6\cdot10^{-10}$. □

C.4 Proof of Corollary 3.11

The method is to compare f_N at a mode with f_N at the integer m' nearest to the predicted place $x\log x+\lambda_{\mathrm{L}}x$. By Theorem 3.9(b) both values are close to values of f_L, and f_L falls off quadratically away from λ_{L}. So a mode cannot be far from m'.

We first prove (a). So we assume (H), $E' < \kappa_L/2$ and $m_h \le N/4$, with E_1 and E' as in the statement.

Let m' be the integer nearest to $x\log x + \lambda_{\mathrm{L}} x$. This number increases in $x \ge 1$ and is at least $\lambda_{\mathrm{L}} > -\frac{1}{2}$, so $m' \ge 0$. Also $|\lambda_{m'} - \lambda_{\mathrm{L}}| \le \frac{1}{2x}$, and since $f_L'(\lambda_{\mathrm{L}}) = 0$, $f_L(\lambda_{m'}) \ge f_L(\lambda_{\mathrm{L}}) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde m$ be a mode of f_N or $\tilde m = m^*$. We claim that $f_N(\tilde m) \ge f_N(m') - 16xH_N/N^2$. For a mode of f_N this is clear. For m^* we have $f_N(m^*) + f_N(N-m^*) = 2P_N(N-m^*) \ge 2P_N(N-m') \ge f_N(m')$. Here $N - m^* \ge \lceil N/2\rceil \ge \lceil N/4\rceil \ge m_h$. So f_N is nonincreasing on $\{\lceil N/4\rceil, \dots, N\}$ by Lemma 3.4. There are at least $N/4$ integers from $\lceil N/4\rceil$ to $\lceil N/2\rceil$. So Lemma B.1 (with $a = \lceil N/4\rceil \ge m_h$ and $b = \lceil N/2\rceil$) gives the second inequality below, and Lemma B.2 gives the third,

$$f_N(N-m^*) \le f_N(\lceil N/2\rceil) \le \frac{\mathbb{P}(M_N \ge \lceil N/4\rceil)}{N/4} \le \frac{16xH_N}{N^2}.$$

Now Theorem 3.9(b), used at $\tilde m$ and at m', gives

$$f_L(\lambda_{\tilde m}) \ge x f_N(\tilde m) - E_1 \ge x f_N(m') - \frac{16x^2H_N}{N^2} - E_1 \ge f_L(\lambda_{m'}) - 2E_1 - \frac{16x^2H_N}{N^2} \ge f_L(\lambda_{\mathrm{L}}) - 2E'.$$

Since $2E' < \kappa_L$, Lemma 3.10(c) gives $(\lambda_{\tilde m} - \lambda_{\mathrm{L}})^2 \le 2E'/\kappa_L$, which is the claim.

Now we prove (b). Let N be large and $\log N \le x \le 2\log N$. Then $x\delta_B$, x/N, $16x^2H_N/N^2$ and $\|f_L''\|_\infty/(8x^2)$ are all at most $\log x/x$, because each is $O(\log^5 N/N)$ or $O(1/\log^2 N)$. So $E' \le C_1 \log x/x$ with an absolute constant C_1. As $\log x/x \le \log\log N/\log N$, for large N we have $E' < \kappa_L/2$. Also (H) holds, $\varepsilon \log N \le 2\log^2 N/N \le \frac{1}{40}$ and $804x\log(x+2) \le N$, so Lemma B.5 gives $m_h \le A_0 x \log^2(x+1) \le N/4$. Hence by (a) $|m^* - x\log x - \lambda_{\mathrm{L}} x| \le x\sqrt{2E'/\kappa_L} \le K\sqrt{x\log x}$ with $K = \sqrt{2C_1/\kappa_L}$, and the same holds for every mode of f_N. At the crossing $x_N = 2\log N - \log\log N + O(1)$ for even N, by Theorem 2.7(b). So $\log N \le x_N \le 2\log N$ for large even N. □

D Proofs for the family estimates

These proofs establish the estimates stated in Appendix B. We retain its notation, including the pathwise construction of the chains.

Proof of Lemma B.1. By Lemma 3.4, $f_N(\ell) \ge f_N(b)$ for $a \le \ell \le b$ in the first case, and $f_N(\ell) \ge f_N(a)$ in the second. Sum over ℓ. □

Proof of Lemma B.2. We use the tree of Lemma 3.1 on $[t]$, with marks only on $v > k_0$. Every odd vertex lies in the family of a marked vertex, so $M_t^{[k_0]} \le \sum_{v \text{ marked}} |\mathcal{T}_v|$. We call a family large if its size is at least z, and small otherwise. If $M_t^{[k_0]} \ge z$, then some marked family is large, or the small marked families have total size at least z. We use the union bound for the first event and Markov's inequality for the second. Marks and tree are independent, so

$$\mathbb{P}\big(M_t^{[k_0]} \ge z\big) \le \varepsilon \sum_v \mathbb{P}(|\mathcal{T}_v| \ge z) + \frac{\varepsilon}{z}\sum_v \mathbb{E}\big[|\mathcal{T}_v|; |\mathcal{T}_v| < z\big].$$

By the first sum in Lemma 2.4, with t in place of N, on average $\frac{t}{d(d+1)}$ vertices have a family of size d. So the first sum over v is $\sum_{d\ge z} \frac{t}{d(d+1)} \le \frac{t}{z}$, and the second is $\sum_{d=1}^{z-1} \frac{t}{d+1} = t(H_z - 1)$. Together they give $\frac{\varepsilon t H_z}{z}$. For the mean, only the vertices $v > k_0$ can be marked and $\mathbb{E}|\mathcal{T}_v| = t/v$, so $\mathbb{E}M_t^{[k_0]} \le \varepsilon \sum_{v=k_0+1}^{t} \frac{t}{v} \le \varepsilon t \log \frac{t}{k_0}$. □

Proof of Lemma B.3. By Lemma 3.3, $\mathbb{P}(|\mathcal{T}_V| = d) = \sum_j \varepsilon(1-\varepsilon)^{j-2}\beta_j(d)$, and $1-(j-2)\varepsilon \le (1-\varepsilon)^{j-2} \le 1$. Apply the two sums of Lemma 2.4, and sum the upper bound over $d' \ge d$. □

Proof of Proposition B.4. We bound the inner sum of Lemma 3.3 from below for each early first flip, namely for $2 \le j \le J := \lfloor N/8 \rfloor$. The terms with $j > J$ are nonnegative and we drop them.

Step 1. One value of j. Fix $2 \le j \le J$ and put $\Gamma_d = d - Y_d + Z^{(j)}_{N-d}$ for $1 \le d \le N-j+1$. The inner sum in Lemma 3.3 is $Q_j(m) = \mathbb{E}S_j$ with $S_j = \sum_d \beta_j(d)\mathbf{1}\{\Gamma_d = m\}$. Let $\mathcal{E}$ be the event $\{Z^{(j)}_{N-1} \le m-1,\ Y_{2m} \le m\}$. We claim that

$$S_j \ge \beta_j(m)\Big(\mathbf{1}_{\mathcal{E}} - \frac{2(j-2)}{N}\,Y_{2m}\Big).$$

Off $\mathcal{E}$ the right side is at most 0 and $S_j \ge 0$. So let $\mathcal{E}$ hold. As d grows by one, Y_d grows by 0 or 1 and $Z^{(j)}_{N-d}$ falls by 0 or 1. So every step of Γ lies in $\{-1,0,1\}$, and Γ cannot jump over m. It starts at $\Gamma_1 = 1 + Z^{(j)}_{N-1} \le m$ (note that $Y_1 = 0$). At $d^\circ = m + Y_{2m} \le 2m$ we have $\Gamma_{d^\circ} \ge d^\circ - Y_{d^\circ} \ge d^\circ - Y_{2m} = m$, since $Z^{(j)} \ge 0$ and Y is nondecreasing. (The index d° is allowed, because $2m + j \le N$.) So $\Gamma_d = m$ for some $d \le d^\circ$, and as β_j is nonincreasing, $S_j \ge \beta_j(d^\circ) = \beta_j(m + Y_{2m})$. It remains to compare $\beta_j(m+y)$ with $\beta_j(m)$ for $0 \le y \le m$. By (5),

$$\frac{\beta_j(m+y)}{\beta_j(m)} = \prod_{i=1}^{y}\Big(1 - \frac{j-2}{N-m-i}\Big) \ge 1 - \frac{2y(j-2)}{N}.$$

Here $N - m - i \ge N/2$, so each fraction is at most $\frac{2(j-2)}{N} \le \frac{1}{4}$, and a product of factors $1-q_i$ with $0 \le q_i \le 1$ is at least $1 - \sum_i q_i$. This proves the claim. Now $\mathbf{1}_{\mathcal{E}} \ge 1 - \mathbf{1}\{Y_{2m} > m\} - \mathbf{1}\{Z^{(j)}_{N-1} \ge m\}$. Taking means in the claim gives

$$Q_j(m) \ge \beta_j(m)\big[1 - \mathbb{P}(Y_{2m} > m) - \mathbb{P}(Z^{(j)}_{N-1} \ge m)\big] - \frac{2(j-2)}{N}\,\mathbb{E}Y_{2m}\,\beta_j(m).$$

Step 2. The chance that Y or Z is large. Lemma B.2 gives

$$\mathbb{P}(Y_{2m} > m) \le 2\varepsilon H_{m+1}, \qquad \mathbb{P}(Z^{(j)}_{N-1} \ge m) \le \frac{xH_m}{m}, \qquad \mathbb{E}Y_{2m} \le 2\varepsilon m \log N.$$

None of these bounds depends on j. Put $p = 2\varepsilon H_{m+1} + xH_m/m$. If $p > 1$, then $e_1'(m) > 1$ and there is nothing to prove. So let $p \le 1$.

Step 3. Sum over j. By Lemma 3.3, $f_N(m) \ge \sum_{j=2}^{J} \mathbb{P}(V=j)\,Q_j(m)$. With Steps 1 and 2 this gives

$$f_N(m) \ge (1-p)\sum_{j=2}^{J}\mathbb{P}(V=j)\,\beta_j(m) - \frac{4\varepsilon m \log N}{N}\sum_{j=2}^{J}\mathbb{P}(V=j)\,(j-2)\,\beta_j(m).$$

In the second sum we use $\mathbb{P}(V=j) \le \varepsilon$ and the second sum in Lemma 2.4. So the second term is at most

$$\frac{4\varepsilon m \log N}{N} \cdot \frac{2x(N-m-1)}{m(m+1)(m+2)} \le \frac{x}{m(m+1)}\,8\varepsilon \log N.$$

The first sum is $\mathbb{P}(|\mathcal{T}_V| = m)$ without the terms $j > J$. So by Lemma B.3 it is at least $\frac{x}{m(m+1)}(1 - \frac{2x}{m+2}) - \varepsilon \sum_{j>J}\beta_j(m)$.

Step 4. The late first flips $j > J$. A family that starts after time $N/8$ rarely reaches size m. By (4), $\beta_j(m) \le \frac{j-1}{N-m} r^{j-1}$ with $r = e^{-a}$ and $a = \frac{m-1}{N-1} \le \frac{1}{4}$. We sum this over

$j-1 \ge J$. With $\sum_{i\ge J} i r^i \le r^J\big(\frac{J}{1-r}+\frac{1}{(1-r)^2}\big)$, $1-e^{-a}\ge a/2$ and $N-m\ge \frac34 N$ we get $\varepsilon\sum_{j>J}\beta_j(m)\le \frac{4\varepsilon}{3N}\big(\frac{2J}{a}+\frac{4}{a^2}\big)e^{-aJ}$. Since $\lfloor N/8\rfloor \ge \frac{N-1}{9}$ for $N\ge 64$, we have $aJ\ge \frac{m-1}{9}$. Also $J\le N/8$ and $a\ge \frac{m-1}{N}$. So

$$\varepsilon\sum_{j>J}\beta_j(m)\le x\Big(\frac{1}{3(m-1)}+\frac{16}{3(m-1)^2}\Big)e^{-(m-1)/9}\le \frac{x}{m(m+1)}\,20me^{-m/9}.$$

The last step uses $m+1\le 3(m-1)$ and $m+1\le 3(m-1)^2$, which hold for $m\ge 2$, and $e^{1/9}\le\frac98$, since $17\cdot\frac98\le 20$.

Step 5. Collect. For $0\le p\le 1$ and $q\ge 0$ we have $(1-p)(1-q)\ge 1-p-q$. So Steps 3 and 4 give

$$f_N(m)\ge \frac{x}{m(m+1)}\Big(1-p-\frac{2x}{m+2}-20me^{-m/9}-8\varepsilon\log N\Big),$$

which is the bound with $e_1'(m)$. For $e_1'\le e_1$ we use $\log(m+1)\ge\log 3>1$ and $\log N\ge\log 64>4$. So $H_m\le 1+\log m\le 2\log(m+1)$, $H_{m+1}\le 1+\log N\le\frac54\log N$ and $\frac{2x}{m+2}\le\frac{2x\log(m+1)}{m}$. Hence $e_1'(m)\le\frac{4x\log(m+1)}{m}+\frac{21}{2}\varepsilon\log N+20me^{-m/9}\le e_1(m)$. □

Proof of Lemma B.5. We pick a point ℓ of order $x\log x$ at which $f_N(\ell)$ is at least half of $\frac{x}{\ell(\ell+1)}$. Up to its mode f_N is nondecreasing. So if the mode were far to the right of ℓ, the bins from ℓ to the mode would carry more mass than the tail $\mathbb{P}(M_N\ge\ell)$ has.

Put $L_2=\log(x+2)\ge\log 3>1$ and $\ell=\lceil 200xL_2\rceil$, so that $200\le\ell\le 201xL_2$ and $4\ell\le 804xL_2\le N$. Since $\log N\ge 4$, we have $\varepsilon\le\frac{1}{160}$. We bound the five terms of $e_1'(\ell)$ in Proposition B.4. As $(1+\log s)/s$ decreases for $s\ge 1$, and $\log 200\le 6$, $\log x\le L_2$ and $\log L_2\le L_2$, we get $\frac{xH_\ell}{\ell}\le\frac{x(1+\log(200xL_2))}{200xL_2}\le\frac{7+2L_2}{200L_2}\le\frac{9}{200}$. Next $\frac{2x}{\ell+2}\le\frac{1}{100}$, $2\varepsilon H_{\ell+1}\le 2\varepsilon(1+\log N)\le\frac{1}{80}+\frac{1}{20}$ and $8\varepsilon\log N\le\frac15$. Finally $se^{-s/9}$ decreases for $s\ge 9$, so $20\ell e^{-\ell/9}\le 4000e^{-200/9}<10^{-5}$. The five terms add up to less than $\frac12$, so $f_N(\ell)\ge\frac{x}{2\ell(\ell+1)}$. If $m_h\ge\ell$, Lemmas B.1 (with $a=\ell$, $b=m_h$) and B.2 give $(m_h-\ell+1)f_N(\ell)\le\mathbb{P}(M_N\ge\ell)\le xH_\ell/\ell$. So $m_h\le(\ell+1)(1+2H_\ell)$ in every case. Here $\ell+1\le 202xL_2$, because $xL_2\ge 1$. Also $H_\ell\le 1+\log(201xL_2)\le 7+2L_2\le 9L_2$, so $1+2H_\ell\le 19L_2$. Hence $m_h\le 3838\,xL_2^2$. Since $(x+1)^2\ge x+2$, we have $L_2\le 2\log(x+1)$, and so $m_h\le 15352\,x\log^2(x+1)\le A_0x\log^2(x+1)$. The last claim holds because $804x\log(x+2)\le 4A_0x\log(x+2)$. □

E Proofs for the second difference

We keep the notation of Section 3. This section proves the results of Section 3.5 on the signs of the second difference.

E.1 Proof of Lemma 3.14

We insert the numbers η_i into ζ one at a time. If $\eta_i=0$, the sign pattern does not change. If $\eta_i>0$, then $\zeta_i>0$ or $\zeta_{i+1}>0$, so η_i is placed next to an entry of its own sign, and the pattern does not change. The case $\eta_i<0$ is the same. This proves the first claim. Deleting entries from a list replaces its sign pattern by a subsequence, and η is obtained from w by deleting the entries ζ_i. This proves the second claim. □

E.2 Proof of Lemma 3.15

Put $\xi_t = P_t(1)/P_t(0)$. By (1), $P_{t+1}(0) = (1-\varepsilon)P_t(0)$ and $P_{t+1}(1) = (1-\varepsilon-\frac{c}{t})P_t(1) + \varepsilon P_t(0)$. So

$$\xi_{t+1} - \xi_t = \frac{1}{1-\varepsilon}\Big(\varepsilon - \frac{c\xi_t}{t}\Big), \qquad \xi_2 = \frac{2\varepsilon}{1-\varepsilon}.$$

We claim that $\xi_t < \varepsilon t/c$ for $t \geq 2$. For $t = 2$ this holds because $c < 1-\varepsilon$. If it holds for t, then $0 < \xi_{t+1} - \xi_t \leq \frac{\varepsilon}{1-\varepsilon} < \frac{\varepsilon}{c}$, and it holds for $t+1$. So ξ_t is strictly increasing for $t \geq 2$, and $\xi_t < 1$ whenever $t \leq c/\varepsilon$.

Now (2) with $a = 1$, and $D_t(0) = P_t(1) - 2P_t(0)$, give for $t \geq 2$

$$s_{t+1} = \big(1 - r_t(2)\big)s_t + \sigma_t, \qquad \sigma_t = \Big(\varepsilon - \frac{c}{t}\Big)(\xi_t - 2)\, P_t(0).$$

Here $1 - r_t(2) \geq 1 - \varepsilon - c = \varepsilon > 0$. If $t < c/\varepsilon$, both factors of σ_t are negative and $\sigma_t > 0$. If $t > c/\varepsilon$, the first factor is positive, and the second is first negative and then positive, because ξ_t increases. So $\mathrm{sp}(\sigma_2, \sigma_3, \dots) \preceq \Pi_3$. Let $\Phi_t = \prod_{u=2}^{t-1}(1 - r_u(2)) > 0$. Then

$$\frac{s_t}{\Phi_t} = s_2 + \sum_{u=2}^{t-1} \frac{\sigma_u}{\Phi_{u+1}}, \qquad s_2 = 1 - 3\varepsilon.$$

If $\varepsilon < \frac{1}{3}$, the terms $s_2, \sigma_2, \sigma_3, \dots$ are first nonnegative, then nonpositive, then nonnegative. Their partial sums are nonnegative and nondecreasing in the first stretch, nonincreasing in the second, and nondecreasing in the third. So the sign pattern of the partial sums is a subsequence of Π_3. If $\varepsilon \geq \frac{1}{3}$, then $c/\varepsilon \leq 1$, so the terms are first nonpositive and then nonnegative, and the sign pattern of the partial sums is a subsequence of $(-,+)$. This proves the first claim.

For the second claim let $0 \leq \varepsilon \leq 1$. The proofs of (1) and (2) use only that step $t+1$ is $+1$ with probability $r_t(a)$ given $A_t = a$, so both hold for this ε. We compute $P_3(0) = \frac{1}{2}(1-\varepsilon)^2$ and $P_3(1) = P_3(2) = \frac{1}{2} - P_3(0)$, so $s_3 = P_3(0) - P_3(1) = (1-\varepsilon)^2 - \frac{1}{2}$. Suppose that $s_3 = 0$. Then $\varepsilon = 1 - 2^{-1/2}$ and $P_3(0) = P_3(1) = \frac{1}{4}$. By (2) with $t = 3$ and $a = 1$, $s_4 = r_3(-1)D_3(0) = (\varepsilon - \frac{c}{3})(P_3(1) - 2P_3(0)) = -\frac{5\varepsilon - 1}{12}$. This is not 0, because $\varepsilon \neq \frac{1}{5}$. (With P_t defined by (1), the same computation works for every real ε, because the other root $1 + 2^{-1/2}$ of s_3 is not $\frac{1}{5}$ either. We need only $0 < \varepsilon < \frac{1}{2}$.) □

E.3 Proof of Theorem 3.16

For $3 \leq t \leq N$ let $\mathcal{Z}_t$ be the list

$$\mathcal{Z}_t = \big(s_N, s_{N-1}, \dots, s_{t+1},\ s_t, D_t(2), D_t(3), \dots, D_t(t-2), s_t,\ s_{t+1}, \dots, s_{N-1}, s_N\big).$$

It has $2N - t - 1$ entries. Its middle part is $(D_t(1), D_t(2), \dots, D_t(t-1))$, because $D_t(t-1) = D_t(1) = s_t$ by symmetry. The list $\mathcal{Z}_3$ is $(s_N, \dots, s_4, s_3, s_3, s_4, \dots, s_N)$, and $\mathcal{Z}_N$ is the list in the statement.

We check that $\mathcal{Z}_{t+1}$ is obtained from $\mathcal{Z}_t$ as in Lemma 3.14, for $3 \leq t \leq N-1$. The first $N - t$ entries of $\mathcal{Z}_{t+1}$ are $s_N, \dots, s_{t+1}$, and they are copies of the first $N - t$ entries of $\mathcal{Z}_t$. So $A_i = 1$ and $B_i = 0$ there. The next $t - 2$ entries are $D_{t+1}(b)$ for $2 \leq b \leq t-1$. By (2),

$$D_{t+1}(b) = r_t(b-2)\, D_t(b-1) + \big(1 - r_t(b+1)\big)D_t(b),$$

and $D_t(b-1)$ and $D_t(b)$ are neighbors in the middle part of $\mathcal{Z}_t$, at the right places. The coefficients satisfy $r_t(b-2) \geq \varepsilon$ and $1 - r_t(b+1) \geq 1 - \varepsilon - c = \varepsilon$, since $2 \leq b \leq t-1$. The last $N-t$ entries of $\mathcal{Z}_{t+1}$ are $s_{t+1}, \ldots, s_N$, and each is a copy of the entry of $\mathcal{Z}_t$ one place to its right. So $A_i = 0$ and $B_i = 1$ there.

By Lemma 3.14, $\mathrm{sp}(\mathcal{Z}_N) \preceq \mathrm{sp}(\mathcal{Z}_{N-1}) \preceq \cdots \preceq \mathrm{sp}(\mathcal{Z}_3)$. By Lemma 3.15 the sign pattern p of $(s_3, \ldots, s_N)$ is a subsequence of Π_3, so it is one of the lists $()$, $(+)$, $(-)$, $(+,-)$, $(-,+)$ and $(+,-,+)$. The sign pattern of $\mathcal{Z}_3$ is p reversed, followed by p, with equal neighbors merged. In the six cases this is $()$, $(+)$, $(-)$, $(-,+,-)$, $(+,-,+)$ and $(+,-,+,-,+)$, and each is a subsequence of Π_5.

For f_N we argue in the same way with $D_t^f(m) = f_t(m+1) - 2f_t(m) + f_t(m-1)$ and the lists

$$\mathcal{Z}_t^f = \big(s_N^f, \ldots, s_{t+1}^f,\ s_t^f, D_t^f(2), \ldots, D_t^f(t-1),\ d_t, d_{t+1}, \ldots, d_{N-1}\big), \qquad 2 \leq t \leq N,$$

where $s_t^f = D_t^f(1)$ and $d_u = (1 - r_u(u+1))D_u^f(u) = (\varepsilon - \frac{c}{u})f_u(u-1)$. The entry $D_{t+1}^f(t)$ of $\mathcal{Z}_{t+1}^f$ equals $r_t(t-2)D_t^f(t-1) + d_t$ by (2), and the other entries are as before. The proof of Lemma 3.15 applies to s_t^f without change, with $\xi_t = f_t(1)/f_t(0)$, $\xi_2 = \frac{\varepsilon}{1-\varepsilon}$ and $s_2^f = 1 - 3\varepsilon$. The numbers d_u are negative for $u < c/\varepsilon$ and positive for $u > c/\varepsilon$. So $\mathrm{sp}(\mathcal{Z}_2^f)$ is a subsequence of $(+,-,+)$ followed by $(-,+)$, and $\mathcal{Z}_N^f = (D_N^f(1), \ldots, D_N^f(N-1))$. As before $\mathrm{sp}(\mathcal{Z}_N^f) \preceq \mathrm{sp}(\mathcal{Z}_2^f)$, and a subsequence of $(+,-,+)$ followed by $(-,+)$ is a subsequence of Π_5 once equal neighbors are merged. This argument needs only $N \geq 2$, and for $N = 2$ the lists $\mathcal{Z}_2^f$ and $\mathcal{Z}_N^f$ are the same. □

E.4 Proof of Theorem 3.17

By symmetry $D_N(N - a_1) > 0 > D_N(N - a_0)$. Suppose that $D_N(a) \leq 0$ for some a with $a_1 < a < N - a_1$.

The case $D_N(a) < 0$. Then the entries of $\mathcal{Z}_N$ at $a_0 < a_1 < a < N - a_1 < N - a_0$ have the signs $-,+,-,+,-$. So $(-,+,-,+,-) \preceq \mathrm{sp}(\mathcal{Z}_N) \preceq \Pi_5$ by Theorem 3.16. This is impossible, because a subsequence of Π_5 with five terms is Π_5 itself.

The case $D_N(a) = 0$. We use the lists $\mathcal{Z}_t$ of the proof of Theorem 3.16 and number the entries of $\mathcal{Z}_t$ from 1 to $2N - t - 1$. So $D_t(b)$ stands at place $N - t + b$ for $1 \leq b \leq t-1$. For $3 \leq t \leq N-1$ let W_t be the list w of Lemma 3.14 for $\zeta = \mathcal{Z}_t$ and $\eta = \mathcal{Z}_{t+1}$. Then $\mathrm{sp}(W_t) = \mathrm{sp}(\mathcal{Z}_t) \preceq \Pi_5$. The entry of $\mathcal{Z}_{t+1}$ at place k is a combination of the entries of $\mathcal{Z}_t$ at the places k and $k+1$, with nonnegative coefficients. We call these two entries its parents. For $2 \leq b \leq t-1$ the parents of $D_{t+1}(b)$ are $D_t(b-1)$ and $D_t(b)$, and both coefficients are positive.

For $4 \leq t \leq N$ let B_t be the set of integers b with $\max(2, a - N + t) \leq b \leq \min(a, t-2)$. It is not empty, because $2 \leq a \leq N-2$. We prove by downward induction on t that there are places $p_1 < p_2 < p_4 < p_5$ in $\mathcal{Z}_t$ such that the entries there have the signs $-,+,+,-$, that $D_t(b) = 0$ for every $b \in B_t$, and that all the places $N - t + b$ with $b \in B_t$ lie strictly between p_2 and p_4. For $t = N$ we have $B_N = \{a\}$, and we take the places a_0, a_1, $N - a_1$ and $N - a_0$.

Let the claim hold for $t+1$, where $4 \leq t \leq N-1$, with the places $p_1' < p_2' < p_4' < p_5'$. Let $b \in B_{t+1}$. Then $D_{t+1}(b) = 0$ is a combination of its two parents with positive coefficients. So the parents both vanish, or they have strictly opposite signs. In the second case one parent is negative. In W_t it stands next to $D_{t+1}(b)$, hence between the positive entries of $\mathcal{Z}_{t+1}$ at the places p_2' and p_4', and these stand between the negative entries at p_1' and p_5'. So $(-,+,-,+,-) \preceq \mathrm{sp}(W_t) \preceq \Pi_5$, which is impossible. Hence $D_t(b-1) = D_t(b) = 0$ for every $b \in B_{t+1}$. Every element of B_t is of the form b or $b-1$ with $b \in B_{t+1}$, so D_t vanishes on B_t.

The entry of $\mathcal{Z}_{t+1}$ at p_1' is negative, so one of its parents is negative. Let $p_1 \in \{p_1', p_1' + 1\}$ be its place in $\mathcal{Z}_t$, and choose p_2, p_4 and p_5 in the same way. Then $p_1 \leq p_1' + 1 \leq p_2' \leq p_2$, and

$p_1 \neq p_2$ because the signs differ. In the same way $p_4 < p_5$. Let q^- and q^+ be the smallest and the largest place $N-t-1+b$ with $b \in B_{t+1}$. The places $N-t+b$ with $b \in B_t$ lie between q^- and $q^+ + 1$. Since $p_2 \leq p_2' + 1 \leq q^-$ and $p_4 \geq p_4' \geq q^+ + 1$, and the entries at p_2 and p_4 are not zero, these places lie strictly between p_2 and p_4. This proves the claim for t.

The induction step with $t = 4$ shows that both parents of $D_5(b)$ vanish for every $b \in B_5 \subseteq \{2, 3\}$. The parents of $D_5(2)$ are s_4 and $D_4(2)$, and those of $D_5(3)$ are $D_4(2)$ and s_4. So $s_4 = D_4(2) = 0$. By (2) with $t = 3$ and $a = 2$, and since $D_3(1) = D_3(2) = s_3$, we have $D_4(2) = r_3(0)s_3 + \big(1 - r_3(3)\big)s_3 = 2\varepsilon s_3$. So $s_3 = 0$ as well, which contradicts Lemma 3.15. □

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